The Unique Games Conjecture

Abstract

We prove the Unique Games Conjecture. For every fixed $\varepsilon, \delta \in (0, 1/2)$, we give a deterministic polynomial-time reduction from 3SAT to Unique Games over a fixed finite alphabet, with completeness at least $1 - \varepsilon$ and soundness at most δ. The output graph is simple and bipartite, and its constraints are translations on a binary vector space. The construction combines a nonlinear map stable under a new noise law with linear detection of that noise. Grassmann expansion and Fourier decoding turn high test acceptance into agreement of local strategies, while parallel repetition bounds that agreement on unsatisfiable inputs.

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1 Introduction

A permutation constraint prescribes exactly one label at one endpoint for each label at the other. The Unique Games Conjecture asserts that systems of such constraints can be hard to approximate even when almost all constraints can be satisfied. We resolve this conjecture positively, with an explicit unweighted reduction from 3SAT.

1.1 The theorem

A Unique Games instance has a finite vertex set, a finite alphabet K, and a nonempty list of oriented constraints $e = (u_e, v_e, \pi_e)$, where π_e is a permutation of K. A labeling a satisfies e when $a(v_e) = \pi_e(a(u_e))$. Its value is the fraction of satisfied constraints, and $\mathrm{val}(G)$ is the maximum value of a labeling. The instance is explicit and unweighted when the entire vertex and constraint lists are written out, including the table of each permutation. A simple bipartite instance has two disjoint vertex sets and at most one constraint on each pair of opposite vertices.

Theorem 1.1 (Unique Games Theorem)**.** *For every fixed $\varepsilon, \delta \in (0, 1/2)$, there are an integer $s \geq 1$ and a deterministic polynomial-time reduction from 3SAT to explicit unweighted Unique Games instances G_φ over $K = \mathbb{F}_2^s$ such that*

$$\begin{aligned} \varphi \text{ satisfiable} &\implies \operatorname{val}(G_\varphi) \geq 1 - \varepsilon, \\ \varphi \text{ unsatisfiable} &\implies \operatorname{val}(G_\varphi) \leq \delta. \end{aligned}$$

The graph is simple and bipartite, and every constraint is a translation $a(v_e) = a(u_e) + c_e$ of K. The alphabet, the degree of the running-time polynomial, and its constants depend only on ε, δ.

Thus the result resolves the Unique Games Conjecture positively in its usual edge-value formulation [23, Conjecture 2.5]. The two errors are prescribed independently, before the alphabet is chosen. Binary translation constraints are a known equivalent formulation of the conjecture [24, Corollary 13 and Section 11.1]; the reduction here produces that form directly.

Theorem 1.1 supplies the Unique Games hypothesis in earlier hardness reductions for cut, covering, constraint-satisfaction, ordering, deletion, and clustering problems [24, 27, 36, 17, 9, 10]. The reductions and the resulting approximation thresholds are due to those authors. Section 8 gives the precise problem formulations, strict thresholds, and reduction qualifications, together with proofs of the consequences.

1.2 Proofs, codes, and the quantifiers

The central difficulty is to retain completeness near one when every label permits only one reply. The earlier connection between proof verification and approximation hardness was developed by Feige, Goldwasser, Lovász, Safra, and Szegedy [15], by Arora and Safra through proof composition [3], and by Arora, Lund, Motwani, Sudan, and Szegedy through the constant-query PCP theorem [2]. Bellare, Goldreich, and Sudan introduced the long code and folding for verifier composition [7, Section 3.3]. The long code of an answer a is the table $f \mapsto f(a)$, indexed by Boolean functions on the answer set. In Håstad's parity-hardness proof, Fourier analysis recovers an outer-game strategy from arbitrary proof tables that make his verifier accept sufficiently often, without first recovering nearby codewords. His nearly satisfiable parity gap is our starting hardness input [18].

Khot introduced the Unique Games Conjecture in 2002 and gave an algorithm based on semidefinite rounding [22]. Several algorithmic guarantees explain why its order of quantifiers matters. For a fixed target error, Trevisan combines semidefinite rounding with graph decomposition to obtain a polynomial-time guarantee when the input error shrinks at the reciprocal-logarithmic rate in the instance size [41]. Charikar, Makarychev, and Makarychev use randomized semidefinite rounding to obtain expected value $1 - O(\sqrt{\epsilon \log q})$ from a q-label instance of value at least $1 - \epsilon$ [8]; this guarantee loses accuracy as the alphabet grows, whereas the conjecture permits a large fixed alphabet after the errors are chosen. Kolla's spectral enumeration gives a running-time bound in terms of the dimension of a suitable spectral subspace of the label-extended graph, whose vertices are variable–label pairs [30]. Arora, Barak, and Steurer use spectral decomposition and enumeration to obtain near-satisfying solutions for sufficiently small fixed completeness error, with a guarantee independent of the fixed alphabet size [1]; their running-time bound is subexponential, not polynomial, for fixed positive error. These are distinct limits on the known guarantees.

Integrality gaps concern the strength of a relaxation. Khot and Vishnoi constructed Unique Games with semidefinite value arbitrarily close to one and integral value arbitrarily close to zero, even with triangle inequalities [29]. Barak, Gopalan, Håstad, Meka, Raghavendra, and Steurer introduced the short code, based on Reed–Muller codes, as a shorter substitute for the long code in applications including alphabet reduction and quantitative integrality gaps [5]. These constructions

constrain particular relaxations; a hardness reduction requires a separate argument. Bafna, Barak, Kothari, Schramm, and Steurer obtain polynomial-time constant-value guarantees for nearly satisfiable translation games whose graphs have suitable low-degree sum-of-squares hypercontractivity certificates, including noisy hypercube and noisy Reed–Muller short-code graphs [4]. The rank-one matrix shortcode used below is a different graph.

1.3 Grassmann expansion and the completeness obstacle

The closest preceding hardness breakthrough concerns 2-to-1 and 2-to-2 games with imperfect completeness. A 2-to-1 constraint is a projection with exactly two preimages per right label. A 2-to-2 constraint partitions the labels on each side into pairs and matches these pairs bijectively, allowing the four combinations within a matched pair. Khot, Minzer, and Safra's STOC 2017 work proposed a Grassmann-graph reduction whose nonstandard soundness guarantee was conditional on a structural hypothesis [26]. Dinur, Khot, Kindler, Minzer, and Safra's STOC 2018 work proposed a reduction with ordinary edge-satisfaction soundness, conditional on an agreement hypothesis [12]; their expansion work formulated the relevant structural problem and established partial cases [11]. Barak, Kothari, and Steurer connected expansion to the needed agreement test through a binary matrix representation [6]. The expansion theorem of Khot, Minzer, and Safra completed this line in 2018 [25].

These results established hardness for almost satisfiable 2-to-1 and 2-to-2 games and, as a consequence, a Unique Games gap with completeness close to one half and arbitrarily small soundness. Obtaining Unique Games completeness arbitrarily close to one was a separate problem.[1]

The degree-two matrix shortcode of Barak, Kothari, and Steurer [6] is the formulation used here: its vertices are binary matrices and its steps add rank-one matrices. Their two-reply rule permits either the old answer or its translate by the update's left factor; requiring equality alone gives a one-half completeness guarantee. They discussed higher-degree tests as a possible route to improving this completeness [6, Section 1.6]. The inverse statement we use, derived from Grassmann expansion, appears in Theorem 5.1. Ellis, Kindler, and Lifshitz later gave a shorter expansion proof using hypercontractivity on spaces of linear maps [14]; our external input remains the Khot–Minzer–Safra theorem. Our proof keeps the established degree-two inverse theorem and changes the noise using a nonlinear map. The following comparison explains that choice.

1.4 Why the test has high completeness

The uniform rank-one test illustrates the completeness problem. Let M be a binary ℓ-by-m matrix and fix a nonzero answer $z \in \mathbb{F}_2^m$. The evaluation function $f_z(M) = Mz$ satisfies

$$f_z(M + al^\top) - f_z(M) = a(l^\top z).$$

For independent uniform $a \in \mathbb{F}_2^\ell$ and $l \in \mathbb{F}_2^m$, this evaluation keeps its value with probability $(1+2^{-\ell})/2$, close to one half. Our test changes the noise while retaining the ordinary rank-one test as the structural tool for soundness.

For any prescribed $p > 0$, Lemma 3.1 fixes a rank threshold and, for every sufficiently large ℓ, constructs a binary space $\mathcal{V}$ containing $K = \mathbb{F}_2^\ell$, a map $C : \mathcal{V} \to K$, and a noise law μ on $\mathcal{V}$ such that, for $a \sim \mu$,

$$C(x + h) = C(x) + h \quad (h \in K), \qquad \Pr[C(x + a) \neq C(x)] \leq p$$

[1] A separate companion proves hardness for exact 2-to-1 games with perfect completeness and arbitrarily small fixed soundness, over fixed alphabets [34, Theorem 1.1].

where x is uniform and independent of a. At the same time, every family of linear observations whose restrictions to K have sufficiently large rank detects a with probability at least $1/8$. The rank threshold depends only on p and is chosen before ℓ. The construction uses quadratic blocks over a finite field: their nonlinear outputs become increasingly stable under recursion, while a rank potential controls the loss of linear observations. Section 3 proves this separation and realizes the output translations as the subspace K.

For the same nonzero z, let $P \in \mathrm{Hom}(\mathbb{F}_2^m, \mathcal{V})$ be uniform. The nonlinear evaluation $P \mapsto C(Pz)$ compares the values

$$C(Pz) \quad \text{and} \quad C\big(Pz + a(l^\top z)\big)$$

under the update $P \mapsto P + al^\top$. Here Pz is uniform in $\mathcal{V}$, independently of the noise a, and $l^\top z$ is an independent fair bit. Its equality test therefore fails with probability at most $p/2$.

We start the reduction from binary parity equations on three distinct variables. Each equation has four satisfying triples. A table on an equation tuple $U = (e_1, \ldots, e_k)$ is an affine map from the product of their solution planes to $\mathcal{V}$, represented by a matrix with $m = 1 + 2k$ columns: one intercept column and two slope columns per equation. Whenever one equation block affects a table only through one actual bit, its reduced key records that variable and the entire induced affine map. A projected table and its pullback receive the same key. Grouping constant K-translates of keys into a vertex then folds the answers: a label at the representative determines every translated answer, and a shared key gives the same restored answer on both domains. The resulting Unique Games test checks equality of the restored answers on a table and its rank-one perturbation using the noise. A near-satisfying global parity assignment gives the ideal evaluation above on every tuple it satisfies, supplying near completeness. The key sharing adapts Khot and Safra's virtual tables [28, Section 1.1], while the translation orbits adapt the folding principle of Bellare, Goldreich, and Sudan [7, Section 3.3].

1.5 Soundness and completion

Soundness follows by comparing two bounds on agreement between local strategies. High matrix-test acceptance gives a positive lower bound independent of the tuple length; the NO parity gap gives an upper bound tending to zero. All alphabet and inverse-theorem parameters are fixed before the tuple length k grows along sufficiently large integer cubes.

1. *Recover structure from high acceptance.* Linear detection of the noise forces a fixed amount of Fourier mass at low restriction rank. Lemma 5.4 converts that mass into positive acceptance for the ordinary uniform rank-one matrix test. The inverse theorem (Theorem 5.1) then gives agreement with an affine evaluation on a slice defined by a bounded number of row and column equations.

2. *Turn that structure into local answers.* In the projected advice experiment (Definition 5.2), each strategy sees its own question and linear advice. An erasure argument finds agreeing slices from a positive fraction of the advice; folding gives a short nonempty set of valid answers from each such slice. Comparing the projected and unrestricted table laws lets the first strategy's chosen slice support the analysis of the second strategy's Fourier decoder. The decoder uses only its own advice. Independent sampling from the first strategy's answer set then gives full answer agreement with probability at least a fixed $\gamma > 0$ (Proposition 5.3).

3. *Bound agreement on a NO instance.* The sparse projection leaves many singleton coordinates where all advice slopes vanish. The proof conditions on data that preserve the product law of

the remaining outer questions, then reconstructs each strategy's input from its own questions. Parallel repetition bounds agreement by $\exp(-ck^{1/3})$ for a fixed $c > 0$ (Lemma 6.2). This contradicts the lower bound γ for large k and establishes the matrix-game gap in Proposition 6.3.

4. *Complete the reduction.* Lemma 7.1 rounds the weights and subdivides every constraint occurrence into four edges, producing a simple bipartite graph while preserving a fixed soundness gap. Theorem 6.1 then gives the prescribed soundness by alphabet-independent repetition. Its exponent is chosen before the alphabet, and the matrix-game completeness error is chosen small enough to survive repetition. Section 7 makes these choices and verifies the polynomial encoding.

The sparse question comparison and Fourier decoding adapt Khot and Safra's method [28]; the correlated advice and vanishing coordinates follow Khot, Minzer, and Safra [26, Sections 3.3–3.4], with later use by Minzer and Zheng [31, Section 3].

For repetition we use the explicit projection-game bound of Dinur and Steurer [13, Corollary 1.2]. Raz established exponential decay for general two-prover games [39], and Holenstein simplified the proof [20]; those bounds depend on the answer alphabet. Rao removed that dependence for projection games [38, Theorem 4]. The Dinur–Steurer bound is applied first on the clean coordinates above and finally after rounding the matrix-game weights and subdividing every edge into four edges. Its alphabet independence permits the final repetition exponent to be fixed before the alphabet, completing the explicit unweighted reduction.

Figure 1 displays the reduction and the two incompatible bounds that establish soundness.

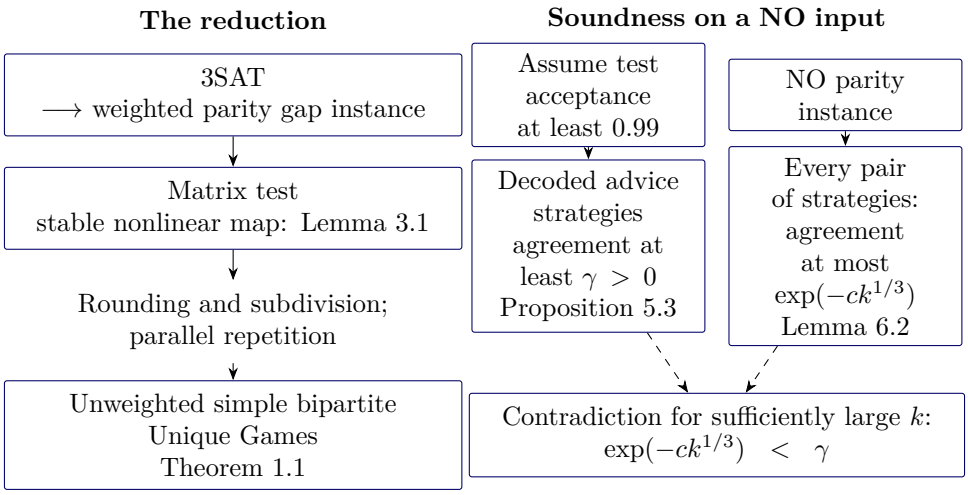


Figure 1: The solid arrows describe the reduction; the dashed arrows describe its soundness argument. The decoding lower bound follows from high test acceptance, while the clean-coordinate upper bound follows from the NO parity gap and applies to every advice strategy. The constants $c, \gamma > 0$ are fixed before the tuple length k grows.

The game and probability conventions are in Section 2. Sections 3 and 4 construct the alphabet and the permutation test used in the roadmap. Section 8 derives the approximation consequences; Appendix A supplies the strengthened left-vertex formulation and finite ordering reduction needed there.

2 Conventions

Weighted games

Intermediate instances may have loops, repeated constraint occurrences, and positive rational weights summing to one. Their value is the maximum total weight satisfied by a labeling. Occurrences retain separate IDs even if their constraints coincide. All rational numbers are encoded by binary numerators and denominators.

A bipartite projection game has a question for each vertex on either side, finite answer sets, and a map from left answers to right answers on each constraint. A strategy chooses one answer to each question. Answer sets that depend on the question can be identified with fixed alphabets of the same size; we use this for the four satisfying triples of a parity equation. Each prover uses only its own question. Private or shared randomness independent of the questions gives a mixture of deterministic strategies and cannot increase value.

The parallel power $G^{\otimes t}$ samples t independent constraint occurrences, sends the ordered question tuples, and accepts if every coordinate accepts. Answers may depend on the prover's entire tuple. Repeated question IDs retain separate coordinate positions and separate local answer copies. This is the ordinary classical product game used by the repetition theorem.

Binary linear algebra and Fourier analysis

Except where a larger field is specified, all vector spaces and maps are finite-dimensional over $\mathbb{F}_2$. Write $\mathrm{Hom}(E, F)$ for the space of linear maps and $E^* = \mathrm{Hom}(E, \mathbb{F}_2)$ for the dual. A subspace $S \le E^*$ restricts to its image $S|_K \le K^*$ on a subspace $K \le E$; its restriction rank is $\dim(S|_K)$.

For a real function f on a binary space E, use normalized Fourier coefficients

$$\widehat{f}(g) = \mathbb{E}_{x \in E} f(x)(-1)^{g(x)}, \qquad f(x) = \sum_{g \in E^*} \widehat{f}(g)(-1)^{g(x)}, \qquad \sum_g |\widehat{f}(g)|^2 = \mathbb{E}_x |f(x)|^2.$$

These identities follow because a nonzero linear functional has equally many zero and one values, so its character has mean zero. The characters are therefore an orthonormal basis. On a space of matrices, this is the entrywise pairing with its dual matrix space; we specify tensor notation when defining the Fourier decoder.

Probability and finite constructions

Uniform sampling from a vector space includes zero. Samples are independent unless a dependence is stated. For finite probability laws,

$$\mathrm{TV}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|.$$

It bounds the difference of expectations of any $[0, 1]$-valued function. We condition only on events of positive probability.

Random experiments describe finite weighted constraint lists. The reduction enumerates those lists deterministically. Every auxiliary dimension and tuple length is fixed before the input formula varies; the final proof checks both this parameter order and the binary encoding cost. Fourier witnesses and decoders are used only to bound game value and need not be efficient.

3 A nonlinear alphabet construction

The construction in this section gives a nonlinear observable that is stable under additive noise, while every family of linear observables whose restrictions to the shift subspace K have sufficiently large rank detects that noise with constant probability. The output translations must also be realized by shifts in an identified subspace of the input. Lemma 3.1 states these requirements together.

Lemma 3.1 (Nonlinear alphabet construction)**.** *For every $p_* > 0$ there is an integer r_* such that, for every sufficiently large integer ℓ, the following objects exist: a finite-dimensional binary space $\mathcal{V}$ containing an identified subspace $K = \mathbb{F}_2^\ell$, a map $C : \mathcal{V} \to K$, and a rational probability distribution μ on $\mathcal{V}$. They satisfy*

$$C(x + k) = C(x) + k \qquad (x \in \mathcal{V},\ k \in K), \tag{3.1}$$

and, for independent uniform $x \in \mathcal{V}$ and $a \sim \mu$,

$$\Pr[C(x + a) \neq C(x)] \leq p_*. \tag{3.2}$$

Moreover, every linear subspace $G \leq \mathcal{V}^*$ with $\dim(G|_K) \geq r_*$ satisfies

$$\Pr_{a\sim\mu}[\text{there exists } g \in G \text{ with } g(a) \neq 0] \geq \frac{1}{8}. \tag{3.3}$$

In particular, r_* is fixed before ℓ is chosen. For rational p_*, the threshold and all the finite data can be found by a terminating deterministic algorithm.

We first construct a gadget with a fixed smaller output space B, in which shifts induce translations of B. After identifying shifts that induce the same translation, this gadget will detect every linear family whose restrictions fill B^*. We then combine copies indexed by injections $B \to K$: a sufficiently high-rank family on K restricts onto all of B^* on most copies. This second step allows every sufficiently large dimension of K while keeping the rank threshold fixed.

The fixed-B construction iterates a quadratic block with two useful properties. On an independent uniform input, a uniform nonzero perturbation in one of its update spaces leaves the nonlinear output unchanged with probability θ. For linear observations, the relevant quantity is the rank left after restriction to that update space. A family satisfying the genericity condition of Definition 3.5 loses at most one dimension, and a uniform choice of update space causes a loss with probability at most $3r\theta$ when the family has rank r. The harmonic potential in (3.9) converts this into an expected loss of at most 3θ per level. Random changes of coordinates make nongeneric families rare as they descend, even when later linear maps depend on earlier choices. Iteration thus makes the nonlinear change probability small while preserving a constant probability of positive rank. A family that reaches the final leaf with positive rank detects an independent uniform symbol there with constant probability.

3.1 The quadratic block

Fix an integer $r_0 \geq 1$, whose size will eventually depend only on p_*. This will be the initial rank tracked in Proposition 3.9; the proof of Lemma 3.1 chooses it large enough to obtain the required nonlinear stability. Write $H_0 = 0$ and $H_r = \sum_{j=1}^{r} 1/j$. Let $F = \mathbb{F}_{2^d}$, $q = 2^d$, and regard $X = B = F^3$ as binary spaces, each of dimension $b = 3d$. The integer d will be chosen sufficiently large in terms of r_0.

Use the trace pairing to identify both X^* and B^* with F^3: the vector $z \in F^3$ represents the character $y \mapsto \mathrm{Tr}(z \cdot y)$, where the dot denotes the field dot product. Here $\mathrm{Tr} : F \to \mathbb{F}_2$ is the field trace. The trace pairing is nondegenerate, and $\mathrm{Tr}(t^2) = \mathrm{Tr}(t)$. Every element of F has a unique square root.

Define

$$Q(x_1, x_2, x_3) = (x_2x_3, x_1x_3, x_1x_2), \qquad D(x, a) = Q(x + a) + Q(x) + Q(a). \tag{3.4}$$

The block observable is

$$C_{\mathrm{blk}} : X \oplus B \longrightarrow B, \qquad C_{\mathrm{blk}}(x, y) = y + Q(x).$$

It is equivariant under the vertical copy of B: $C_{\mathrm{blk}}(x, y+k) = C_{\mathrm{blk}}(x, y)+k$. For $z \in B^*$, the scalar observation $z \circ C_{\mathrm{blk}}$ changes by $z(k)$ under this shift. We shall compare it with linear functionals whose restriction to the vertical B is also z.

For a one-dimensional F-subspace $A = Fv$ of F^3, put

$$D_A = \{y \in B : v \cdot y = 0\}, \qquad U_A = \{(a, Q(a) + w) : a \in A,\ w \in D_A\} \subseteq X \oplus B. \tag{3.5}$$

These definitions are independent of the chosen nonzero generator v. There are $q^2 + q + 1$ such lines; set

$$\theta = \frac{1}{q^2 + q + 1}. \tag{3.6}$$

Lemma 3.2. *For every line A, the space U_A is binary linear of dimension b. For $a \in A$ and $x \in X$, one has $D(x, a) \in D_A$; for $a, a' \in A$, one has $D(a, a') = 0$. The trace annihilator of D_A in $B^* = F^3$ is A, and the spaces U_A together span $X \oplus B$.*

Proof. The three coordinates of $D(x, a)$ are

$$x_2a_3 + a_2x_3, \quad x_1a_3 + a_1x_3, \quad x_1a_2 + a_1x_2.$$

When $a = tv$, every term in $v \cdot D(x, a)$ occurs twice, so their sum is zero. When both arguments are multiples of v, each coordinate of the polar expression is zero. Thus $Q|_A$ is binary additive, and the parametrization of U_A by $A \oplus D_A$ is a binary linear bijection. Its dimension is $d + 2d = b$.

Every $z \in A$ annihilates D_A under the trace pairing. The annihilator has binary dimension $3d - 2d = d$, so it is exactly A. Finally, for each coordinate line $A = Fe_i$, the space U_A contains $(Fe_i, 0)$, because $Q(te_i) = 0$, as well as $\{0\} \oplus D_A$. These three coordinate lines and their vertical hyperplanes span $X \oplus B$. □

The nonlinear response

The form of U_A makes the quadratic term of an update cancel. For $(a, Q(a) + w) \in U_A$,

$$C_{\mathrm{blk}}\big((x, y) + (a, Q(a) + w)\big) + C_{\mathrm{blk}}(x, y) = w + D(x, a).$$

The right-hand side lies in D_A by Lemma 3.2. The following calculation gives the response probability used in the recursion.

Lemma 3.3 (Response to one block update)**.** *Fix a line A. For independent uniform $(x, y) \in X \oplus B$ and $\zeta \in U_A \setminus \{0\}$,*

$$\Pr\big[C_{\mathrm{blk}}((x, y) + \zeta) \neq C_{\mathrm{blk}}(x, y)\big] = 1 - \theta.$$

Proof. First take ζ uniform in all of U_A, still independently of the uniform aggregate. Its parametrization $(a, Q(a) + w)$ has independent uniform $a \in A$ and $w \in D_A$. For every fixed x, a, the output difference $w + D(x, a)$ is uniform in D_A, so it is nonzero with probability $1 - q^{-2}$. The zero perturbation has mass q^{-3} and never changes the output. Conditioning on a nonzero perturbation therefore gives

$$\frac{1 - q^{-2}}{1 - q^{-3}} = \frac{q(q+1)}{q^2 + q + 1} = 1 - \theta.$$

□

Linear restrictions

Fix a binary subspace $S \leq B^*$. A linear functional on $X \oplus B$ whose restriction to the vertical B is $z \in S$ has a unique form $(\gamma_z, z) \in X^* \oplus B^*$. Thus a binary-linear choice of one such extension for each $z \in S$ is specified by a binary linear map $\gamma : S \to X^*$. We ask which of these chosen extensions vanish when restricted to U_A. The restriction loses rank exactly when its kernel contains a nonzero element.

Lemma 3.4 (Kernel of a block restriction)**.** *Let $S \leq B^*$ and let $\gamma : S \to X^*$ be binary linear. For each line A, define*

$$\kappa_{A,\gamma} : S \longrightarrow U_A^*, \qquad \kappa_{A,\gamma}(z) = (\gamma_z, z)|_{U_A}.$$

Under the trace identification, write $\gamma_z(x) = \mathrm{Tr}(g(z) \cdot x)$ for the associated binary linear map $g : S \to F^3$. For every nonzero $z \in S$,

$$\kappa_{A,\gamma}(z) = 0 \quad \Longleftrightarrow \quad A = Fz \quad \textit{and} \quad z \cdot g(z) = \sqrt{z_1 z_2 z_3}.$$

Proof. If z lies in the kernel, it annihilates every vertical vector $(0, w)$ with $w \in D_A$. Lemma 3.2 therefore gives $z \in A$, and for $z \neq 0$ this means $A = Fz$. On the remaining vectors $(tz, Q(tz))$, $t \in F$, the functional has value

$$\mathrm{Tr}(t\, g(z) \cdot z + t^2 z_1 z_2 z_3). \tag{3.7}$$

The three summands of $z \cdot Q(tz)$ are equal, and three equals one in characteristic two. Also

$$\mathrm{Tr}(t^2 c) = \mathrm{Tr}(t\sqrt{c}).$$

Nondegeneracy of the trace pairing shows that (3.7) vanishes for every t exactly when $z \cdot g(z) = \sqrt{z_1 z_2 z_3}$. Together with $z \in A$, this condition is sufficient as well: the vertical part is already annihilated. □

3.2 Generic spaces of characters

The kernel criterion reduces rank loss to two properties of the character space: each field line should contain at most one nonzero character, and few characters should satisfy the alignment equation for any choice of extensions. Requiring the second property for every choice of extensions will allow those extensions to depend on earlier random changes of coordinates.

Definition 3.5. *A binary subspace $S \leq F^3$ of dimension $r \geq 1$ is generic if no two distinct nonzero elements of S are proportional over F, and if, for every binary linear map $g : S \to F^3$, at most $3r$ nonzero $z \in S$ satisfy*

$$z \cdot g(z) = \sqrt{z_1 z_2 z_3}. \tag{3.8}$$

Fix one map $\gamma : S \to X^*$ before choosing a uniform line A, and let $r_A = \dim \operatorname{im} \kappa_{A,\gamma}$, where $r = \dim S > 0$. Lemma 3.4 now bounds the loss. If S is generic, each field line contains at most one nonzero element of S, so the kernel has dimension at most one. For the fixed map g associated with γ, at most $3r$ characters satisfy (3.8); hence at most $3r$ lines can cause any rank loss. Consequently

$$r - r_A \le 1, \qquad \Pr_A[r_A < r] \le 3r\theta, \qquad \mathbb{E}_A[H_r - H_{r_A}] \le 3\theta. \tag{3.9}$$

The last bound uses $H_r - H_{r-1} = 1/r$. For an arbitrary rank-r space, a loss still requires $A = Fz$ for some nonzero $z \in S$. Therefore

$$\Pr_A[r_A < r] \le (2^r - 1)\theta \le 2^r\theta, \qquad H_r - H_{r_A} \le H_r. \tag{3.10}$$

These estimates concern rank under restriction. Detection of the eventual noise will follow only after a positive-rank family reaches a leaf.

It remains to show that spaces with the stated genericity are abundant.

Lemma 3.6. *For each fixed $r \ge 1$, a uniformly random binary r-dimensional subspace of F^3 is generic with probability tending to one as $d \to \infty$.*

Proof. Alignment for a set of characters can be expressed as membership in a column space. We first show that this membership is impossible over a function field when the set is too large. A nonzero minor then bounds the probability of membership after specialization to a finite field.

Random parametrization. Choose independent uniform $x_{ij} \in F$, for $1 \le i \le 3$ and $1 \le j \le r$, and parametrize a binary linear map by

$$z_i(t) = \sum_{j=1}^{r} x_{ij} t_j, \qquad t \in \mathbb{F}_2^r.$$

Each nonzero input has uniform image in F^3, so the probability that this map is not injective is at most $(2^r - 1)q^{-3}$. Conditioned on injectivity, its image is a uniform binary r-subspace: every such subspace has the same number of ordered binary bases. Distinct nonzero binary inputs are linearly independent also over F, so their images are independent uniform elements of F^3. A union bound therefore shows that the nonproportionality condition fails with probability $O_r(q^{-2})$, apart from the event of noninjectivity.

It remains to prove the alignment condition simultaneously for all g. Fix a set E of more than $3r$ distinct nonzero inputs in $\mathbb{F}_2^r$. On an injective parametrization, the values of a binary linear g on a basis can be arbitrary vectors in F^3. Thus alignment on all of E implies that the vector

$$c = \big(\sqrt{z_1(t)z_2(t)z_3(t)}\big)_{t \in E}$$

lies in the F-column span of the matrix with $3r$ columns

$$M_{t,(i,j)} = t_j z_i(t), \qquad t \in E, \quad 1 \le i \le 3, \quad 1 \le j \le r.$$

Conversely, membership supplies the basis values of such a g. In particular, this column-membership condition already accounts for all possible maps g.

The algebraic obstruction. Regard the x_{ij} as independent indeterminates and set $L = \mathbb{F}_2(x_{ij})$. We claim that c is not in the column span of M even after extending scalars to an algebraic closure $\overline{L}$. The following r^3 elements are linearly independent over L:

$$\sqrt{x_{1j}x_{2k}x_{3l}}, \qquad 1 \le j, k, l \le r.$$

Indeed, after clearing denominators, a dependence would be a polynomial identity in the square roots of the indeterminates. The displayed monomials have distinct parity vectors, whereas coefficients polynomial in the x_{ij} have even exponents in those square roots. Terms from different parity vectors cannot cancel.

The expansion of c in these independent elements has coefficient vectors

$$v_{jkl} = (t_j t_k t_l)_{t \in E}.$$

Suppose that c belonged to the column span over $\overline{L}$. Every row vector over L annihilating M would then annihilate c. The independence just proved implies that it would annihilate every v_{jkl}. By double annihilation in the finite-dimensional space L^E, every v_{jkl} would belong to the column span over L itself.

To see why this is impossible, compare the degrees of the functions on E. For $h \geq 1$, let $V_h \leq L^E$ be the evaluation span of all positive-degree monomials in the binary coordinates $t_1, \ldots, t_r$ of degree at most h. Repeated indices in v_{jkl} give all monomials of degrees one and two as well as three, because $t_i^2 = t_i$ on the evaluation set. Each column of M has degree at most two. The preceding conclusion therefore implies

$$V_3 \subseteq \mathrm{col}_L(M) \subseteq V_2 \subseteq V_3.$$

These three spaces are equal. Multiplication by any t_i sends V_2 into V_3, so preserves their common value. Since this space contains the degree-one monomials, it contains evaluations of every positive-degree monomial. These evaluations span all of L^E: for any nonzero binary point, its usual indicator polynomial is a product of factors t_i and $1 + t_i$ with at least one factor t_i, and thus has no constant monomial. Its restriction to E gives the corresponding point indicator. Consequently the column span would have dimension $|E| > 3r$, a contradiction.

Specialization to finite fields. Squaring the entries of both M and c removes the square roots. The resulting matrices have polynomial entries, and the appended column has entries $z_1(t)z_2(t)z_3(t)$. Let $h = \mathrm{rank}_L M \leq 3r$. Squaring preserves rank over $\overline{L}$, so the augmented squared matrix has a nonzero minor of size $h + 1$. This minor is a nonzero polynomial of degree at most $2h + 3 \leq 6r + 3$. All unaugmented minors of size $h + 1$ vanish identically, so the unaugmented rank cannot increase on specialization. It follows that specialized column membership forces this particular nonzero polynomial to vanish. For uniform independent $x_{ij} \in F$, this event has probability at most $(6r+3)/q$ by the polynomial zero bound [40, Corollary 1]; see also the earlier sparse-polynomial formulation in [42, Section 3.1].

If the alignment condition fails, it fails on some set E of size $3r + 1$. There are finitely many such sets depending only on r; if $2^r - 1 \leq 3r$ there are none and the condition is automatic. A union bound, followed by conditioning on injectivity, proves the lemma. □

We choose the field so that nongeneric spaces are rare at every rank up to r_0. Since genericity controls every map g, this field choice will apply to all later choices of extensions. Choose a positive ε_0 with

$$H_{r_0} 2^{r_0} \varepsilon_0 \leq 1, \qquad \varepsilon_0 < 1. \tag{3.11}$$

By Lemma 3.6, we may fix d sufficiently large that

$$b = 3d \geq r_0, \qquad \Pr[S \text{ is nongeneric}] \leq \varepsilon_0 \quad (1 \leq r \leq r_0), \tag{3.12}$$

where S is a uniformly chosen rank-r subspace of B^*. In particular, there exists a generic rank-r_0 subspace, which we fix once and for all.

3.3 Concatenation and its noise

We keep the output space B fixed while concatenating the quadratic blocks. One noisy branch will be chosen at each level. The next lemma computes how fast the nonlinear change probability decays; we will then track the rank of linear restrictions along the same branch.

We construct binary input spaces P_t, shift subspaces $R_t \leq P_t$, linear surjections $\lambda_t : R_t \to B$, and maps $C_t : P_t \to B$ such that

$$C_t(u + h) = C_t(u) + \lambda_t(h) \qquad (h \in R_t). \tag{3.13}$$

A shift $h \in R_t$ thus represents the output translation $\lambda_t(h)$; different shifts may represent the same translation. This form of the shift action is preserved by aggregation. At the end, quotienting by the shifts of value zero will identify the output translations with a subspace of the input.

At height zero take $P_0 = R_0 = B$ and $C_0 = \lambda_0 = \mathrm{id}_B$. Let $\mathcal{I}$ consist of all pairs (A, J), where A is a line in F^3 and $J : B \to U_A$ is a binary linear isomorphism. Each line has the same number of such isomorphisms.

We include one child for every (A, J) in a single fixed parent map. Choosing a uniform noisy child will then sample a uniform line and a fresh uniform isomorphism. For $t \geq 1$, take $P_t = \prod_{(A,J)\in\mathcal{I}} P_{t-1}$ and set

$$(x, y) = \sum_{A,J} JC_{t-1}(u_{AJ}), \qquad C_t(u) = C_{\mathrm{blk}}(x, y) = y + Q(x). \tag{3.14}$$

A tuple of child shifts changes this aggregate by the sum of their embedded output translations. To make the parent change a pure output translation, we require the X-component of this sum to vanish. Accordingly, define

$$R_t = \left\{ (h_{AJ}) \in \prod_{A,J} R_{t-1} : \sum_{A,J} J\lambda_{t-1}(h_{AJ}) \in \{0\} \oplus B \right\}, \tag{3.15}$$

and let $\lambda_t(h) = k$ when this aggregate is $(0, k)$. These are linear definitions, and (3.13) follows immediately from (3.14). Surjectivity follows inductively from Lemma 3.2: the child output translations can be chosen arbitrarily, and their aggregate map is onto $X \oplus B$. The relation (3.13) and surjectivity also show that C_t sends a uniform input to a uniform element of B.

Let ξ_t be the following additive noise in P_t. At height zero it is uniform in B. At a positive height, choose a uniform child (A, J) and embed an independent copy of ξ_{t-1} in that child, with zero in all other children. Equivalently, choose a uniform leaf in the product tree and add a uniform symbol of B at that leaf. Each path choice can be sampled as a uniform A followed by a uniform J, independently at successive levels. All these distributions are rational. Figure 2 illustrates one level of the recursion and the selected noisy branch.

Lemma 3.7. *For independent uniform $u \in P_t$ and noise ξ_t,*

$$p_t := \Pr[C_t(u + \xi_t) \neq C_t(u)] = (1 - q^{-3})(1 - \theta)^t. \tag{3.16}$$

Proof. Shift equivariance makes the initial aggregate uniform even after conditioning on the child output difference. The random isomorphism then makes a nonzero difference uniform in the selected update space, so Lemma 3.3 gives the recursion.

The identity at height zero is immediate. At one higher level, condition on the selected child (A, J). Write Δ for its output difference. All child copies have the same function and the same inner noise law, so the law of Δ is independent of the selected index.

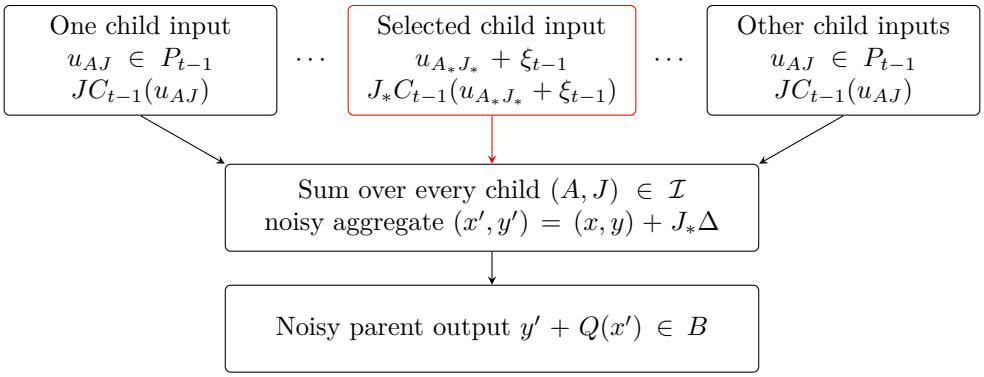


Figure 2: One level of the nonlinear recursion, shown under its additive noise. The input space is a product over all child indices (A, J); only one uniformly chosen child is perturbed. Each child output is embedded in $U_A \leq X \oplus B$, all embedded outputs are added, and the quadratic map is applied once at the parent. Here (x, y) is the initial aggregate, and Δ is the selected child's output difference, so the noisy aggregate is $(x', y') = (x, y) + J_*\Delta$. The omitted children obey the same rule. The highlighted perturbation may disappear at the parent.

We need also the conditional law of the child's initial output. For any fixed child perturbation a, (3.13) gives

$$C_{t-1}(u + a + h) + C_{t-1}(u + h) = C_{t-1}(u + a) + C_{t-1}(u) \qquad (h \in R_{t-1}).$$

Thus the difference is invariant under these shifts, while the initial output is translated by every element of B. On uniform input its initial output remains uniform conditional on the difference, even if the inner perturbation is also specified. All other initial child outputs remain independently uniform. Because the aggregate map is onto $X \oplus B$, the initial aggregate (x, y) is therefore uniform conditional on (A, J, Δ). In particular it is independent of the aggregate perturbation $J\Delta$.

Conditional on $\Delta \neq 0$ and A, the isomorphism J remains uniform, because the law of Δ is independent of the selected child index. Thus $J\Delta$ is uniform in $U_A \setminus \{0\}$. The conditional uniformity just proved makes the initial aggregate independent of this update. Lemma 3.3 therefore gives parent change probability $1 - \theta$ conditional on a nonzero child difference. When $\Delta = 0$, the aggregate does not change. Consequently $p_t = p_{t-1}(1 - \theta)$, proving the identity. □

3.4 Linear lifts and loss of rank

The nonlinear change probability now decays with the height. For the fixed-B gadget, our remaining target is detection whenever a family's restrictions fill B^*. We will prove this by tracking every binary-linear choice of extensions of these output characters through the shift action.

For $S \leq B^*$, a lift on P_t means a binary linear map $z \mapsto \psi_z$ from S to P_t^* satisfying

$$\psi_z|_{R_t} = z \circ \lambda_t. \tag{3.17}$$

A lift is full when its domain is B^*. The next lemma describes the restrictions of an arbitrary parent lift to its children.

Lemma 3.8. *Let a lift with domain S be given on P_t, where $t \geq 1$. For each child (A, J), restriction to that child's shift space factors through λ_{t-1}. The resulting characters on the child output space B have the form*

$$\phi_{AJ}(z) = J^*\big((\gamma_z, z)|_{U_A}\big) \in B^*, \tag{3.18}$$

where $\gamma : S \to X^*$ is a single binary linear map common to all the children.

Proof. A shift supported on one child and lying in $\ker \lambda_{t-1}$ is a parent shift with output translation zero. Each ψ_z vanishes on it, proving the first assertion.

Let $T = \prod_{A,J} B$ be the space of child output translations, with aggregate map $\mathcal{A} : T \to X \oplus B$. This map is onto. The sum of the child characters, minus z applied to the second component of $\mathcal{A}$, vanishes whenever the first component of $\mathcal{A}$ is zero: every such tuple of translations can be supplied by child shifts, giving a parent shift. The difference therefore factors uniquely through the first component of $\mathcal{A}$, which is onto X. Call this character γ_z. Uniqueness shows that γ_z depends linearly on z. Evaluating on a tuple supported at one child gives (3.18). □

After choosing a child, let $S' = \operatorname{im} \phi_{AJ}$ and choose a linear section $\iota : S' \to S$ of ϕ_{AJ}. Restricting $\psi_{\iota(z')}$ to that child input gives a lift with domain S' on P_{t-1}. We may fix deterministic rules for all such sections. In particular they depend only on the already chosen path, and never on a future path index or on the final noise symbol. On descending the tree, every surviving functional is a restriction of a member of the original linear family.

The factorization now connects the recursion to the local restriction problem. For a fixed parent lift, its single map γ is common to every outgoing line A, and $\phi_{AJ} = J^* \kappa_{A,\gamma}$. Since J^* is an isomorphism, it does not change the kernel or rank. We may therefore apply (3.9) and (3.10) when the outgoing line is chosen uniformly. The next proof shows that the incoming isomorphisms make genericity typical at each positive subsequent rank, even though the lift itself depends on the earlier path.

Proposition 3.9. *For the quadratic block of Section 3.1, assume* (3.11) *and* (3.12)*, and fix a generic rank-r_0 subspace of B^*. Let P_t, ξ_t, and full lifts be as constructed in Sections 3.3 and 3.4. Set*

$$s = \left\lfloor \frac{H_{r_0}}{8\theta} \right\rfloor. \tag{3.19}$$

Every full lift on P_s detects ξ_s with probability at least $1/4$*:*

$$\Pr[\textit{there exists } z \in B^* \textit{ with } \psi_z(\xi_s) \neq 0] \geq \frac{1}{4}. \tag{3.20}$$

Proof. We bound the expected loss of harmonic potential along the random path. A sufficiently small total loss gives a positive probability that some rank survives to the leaf, where a uniform noise symbol is detected with probability at least one half.

The conditional law along the path. The full lift at the top restricts to the fixed generic subspace of dimension r_0. Along the random leaf path, each chosen child inherits the image lift described above. Let S_j be its domain after j steps, and let $r_j = \dim S_j$, so initially $r_j = r_0$ at $j = 0$. After extinction the domain remains the zero space.

The order of conditioning makes genericity available at each level. Let $\mathcal{F}_j$ be the information in all path choices before the outgoing choice (A_j, J_j) at step j. The current space, lift, and map γ_j are $\mathcal{F}_j$-measurable. Conditional on $\mathcal{F}_j$ and A_j, the subspace

$$E_{j,A_j} = \operatorname{im}[z \mapsto (\gamma_{j,z}, z)|_{U_{A_j}}] \leq U_{A_j}^*$$

is fixed. The next domain and its rank are

$$S_{j+1} = J_j^* E_{j,A_j}, \qquad r_{j+1} = \dim E_{j,A_j}.$$

The rank is fixed before J_j is revealed. A uniform binary isomorphism $J_j : B \to U_{A_j}$ makes S_{j+1} a uniformly random subspace of B^* of that rank. Therefore, if B_j is the indicator that $r_j > 0$ and S_j is nongeneric, then

$$\mathbb{E}[B_{j+1} \mid \mathcal{F}_j, A_j] \le \varepsilon_0. \tag{3.21}$$

Genericity bounds every binary linear map g on the selected subspace, including the map supplied by the next lift, which may depend on J_j.

Expected loss of harmonic potential. Let $\Delta_j = H_{r_j} - H_{r_{j+1}} \ge 0$. Conditional on $\mathcal{F}_j$, the outgoing A_j is independent and uniform. For a generic current domain, (3.9) bounds the expected loss by 3θ. For a nongeneric domain, (3.10) bounds it by the harmonic potential times the probability of rank loss. Since the rank is at most r_0, these bounds give

$$\mathbb{E}[\Delta_j \mid \mathcal{F}_j] \le 3\theta + H_{r_0} 2^{r_0} \theta B_j. \tag{3.22}$$

For $j = 0$ the space is generic, so its expected drop is at most 3θ. For $j \ge 1$, averaging (3.22) over the preceding isomorphism replaces the nongeneric indicator by its expectation, bounded by ε_0 in (3.21). The choice (3.11) then gives the unconditional bound

$$\mathbb{E}\Delta_j \le (3 + H_{r_0} 2^{r_0} \varepsilon_0)\theta \le 4\theta.$$

Survival and leaf detection. Summing the expected drops and using (3.19),

$$\mathbb{E}[H_{r_0} - H_{r_s}] \le 4s\theta \le \frac{H_{r_0}}{2}.$$

Extinction requires a total drop of H_{r_0}, so $\Pr[r_s = 0] \le 1/2$. At a leaf, $P_0 = R_0 = B$ and its lift consists of the characters themselves. Conditional on any surviving rank $r_s > 0$, an independent uniform symbol in B is detected with probability $1 - 2^{-r_s} \ge 1/2$. Every surviving functional is a restriction of the original family along the selected leaf, so such a detection implies a detection by that family of the embedded noise ξ_s. Multiplying the two bounds proves (3.20). □

3.5 Quotients and enlargement

The fixed-B construction is complete once its shift action is realized by a subspace. Put

$$V_0 = P_s / \ker \lambda_s,$$

where the kernel is taken inside $R_s \le P_s$, and let $q_0 : P_s \to V_0$ be the quotient map. The relation (3.13) makes C_s invariant under this kernel, so it descends to a map $\overline{C} : V_0 \to B$. The quotient $R_s / \ker \lambda_s$ is identified with B by λ_s, giving an inclusion $i_B : B \to V_0$ such that

$$q_0(h) = i_B(\lambda_s h) \qquad (h \in R_s).$$

We henceforth identify this subspace with B. Its shift action satisfies

$$\overline{C}(x + b') = \overline{C}(x) + b' \qquad (b' \in B).$$

Let ν be the pushforward of the noise law of ξ_s. Uniform input on P_s pushes to uniform input on V_0, so the output change probability remains p_s.

Every linear subspace $G_0 \le V_0^*$ whose restriction is all of B^* satisfies

$$\Pr_{a \sim \nu}[\text{there exists } g \in G_0 \text{ with } g(a) \ne 0] \ge \frac{1}{4}. \tag{3.23}$$

Indeed, choose a linear section $\sigma : B^* \to G_0$ of restriction and define $\psi_z = q_0^* \sigma(z) = \sigma(z) \circ q_0$. For $h \in R_s$,

$$\psi_z(h) = \sigma(z)(q_0 h) = z(\lambda_s h).$$

Thus this pullback is a full lift. Moreover, $\psi_z(\xi_s) = \sigma(z)(q_0 \xi_s)$, so detection by the lift is detection by the chosen section on the pushed-forward noise. Proposition 3.9 gives the bound, and detection by the section implies detection by G_0.

Proof of Lemma 3.1. The quotient above gives a gadget with output space B and detection for families that restrict onto all of B^*. We choose its parameters to obtain the desired stability, then combine copies to obtain arbitrary sufficiently large output dimension and the high-rank detection condition.

Parameter choices. Choose r_0 sufficiently large that

$$\exp(1 - H_{r_0}/8) \le p_*.$$

Then choose ε_0 as in (3.11) and fix d, and hence $b = 3d$ and θ, using Lemma 3.6. Construct the height s in (3.19) and the quotient $(V_0, B, \overline{C}, \nu)$ above. Lemma 3.7 gives

$$p_s \le (1-\theta)^s \le \exp(-\theta s) \le \exp(\theta - H_{r_0}/8) \le p_*. \tag{3.24}$$

Set $r_* = b + 2$. All choices so far depend only on p_*.

Enlarging the output space. A family of restriction rank at least $b + 2$ on K restricts onto all of B^* along most injections $B \to K$, as the estimate below will show. A copy of the base gadget for each injection therefore converts its full-restriction detection bound into the required high-rank bound.

Fix $\ell \ge b+2$ and write $K = \mathbb{F}_2^\ell$. Let $\mathcal{L}$ be the finite set of all binary linear injections $L : B \to K$. Take a product $P = \prod_{L \in \mathcal{L}} V_0$ and define

$$\widetilde{C}((x_L)_L) = \sum_{L \in \mathcal{L}} L\overline{C}(x_L).$$

The product of shift subspaces $T = \prod_L B \le P$ acts on this output through the linear map

$$\Lambda : T \longrightarrow K, \qquad \Lambda((b_L)_L) = \sum_L L b_L.$$

This map is onto, because every nonzero vector of K is in the image of some injection $B \to K$. The final input space is the quotient by its kernel inside T:

$$\mathcal{V} = P / \ker \Lambda.$$

The map $\widetilde{C}$ descends to $C : \mathcal{V} \to K$, and $T/\ker\Lambda$ identifies with the required shift subspace K. This proves (3.1).

On P, choose a uniform $L \in \mathcal{L}$ and update only that copy by an independent sample from ν. Let μ be the resulting law after the quotient. On uniform product input the output sum changes exactly when the selected copy's output changes, because L is injective. Uniform input pushes to uniform input on the quotient, so the change probability is still p_s. The relation (3.24) proves (3.2). Every noise distribution used here is obtained from uniform finite choices by pushforward, and therefore is rational.

Detection by high-rank families. Let $G \le \mathcal{V}^*$ and let $S = G|_K \le K^*$ have dimension $r \ge r_*$. The pullback of G to P, restricted to the copy of V_0 indexed by L, gives a subspace $G_L \le V_0^*$. Its restriction to that copy's shift subspace B is exactly

$$G_L|_B = L^*S.$$

To see this, a shift $b' \in B$ supported on copy L represents the shift $Lb' \in K$ in the quotient, so a character restricting to $z \in S$ evaluates there as $z(Lb')$.

The map $L^* : S \to B^*$ fails to be onto if and only if there is a nonzero $b' \in B$ with $Lb' \in S^\perp$. For each such fixed b', a uniform injection L makes Lb' uniform among the nonzero vectors of K. Hence

$$\Pr_L[Lb' \in S^\perp] = \frac{2^{\ell-r}-1}{2^\ell - 1} \le 2^{-r}.$$

A union bound gives

$$\Pr_L[L^*S \ne B^*] \le (2^b - 1)2^{-r} \le 2^{b-r} \le \frac{1}{4}.$$

On each of the remaining copies, (3.23) gives detection probability at least $1/4$ by G_L, and thus by G on the embedded noise. Averaging over the selected copy yields detection probability at least $\frac{3}{4} \cdot \frac{1}{4} = \frac{3}{16} \ge \frac{1}{8}$, proving (3.3). The construction works for every $\ell \ge b + 2$, with the already fixed $r_* = b + 2$, as required.

Effective construction. All choices are effective when p_* is rational. Choose an integer $m \ge 0$ with $2^{-m} \le p_*$ and then choose r_0 with $H_{r_0} \ge 8(m+1)$; since $e^{-m} \le 2^{-m}$, this ensures the initial bound on r_0. Choose rational ε_0 satisfying (3.11). For successive integers d with $3d \ge r_0$, enumerate the binary subspaces of $\mathbb{F}_{2^d}^3$ and the linear maps in the definition of genericity. This tests genericity exactly and computes the fraction of nongeneric spaces at each rank $1, \dots, r_0$. Lemma 3.6 ensures that eventually all these fractions are at most ε_0. Fix the first such field and a generic rank-r_0 space. The remaining products, isomorphisms, injections, quotients, function tables, and rational noise laws are obtained by finite enumeration and binary linear algebra. This gives the asserted terminating construction. □

4 The outer questions and the permutation test

The gadget of Lemma 3.1 supplies the alphabet and noise for a permutation instance built from affine tables on tuples of parity equations. To compare tables on an equation tuple and on a projected tuple, we reduce each table to the coordinates on which it depends, preserving the induced affine map and its intercept. We then identify constant alphabet shifts to obtain the vertices of the instance. This representation gives the shared table answers used in Section 5.

4.1 Weighted parity equations and distinct positions

A *weighted parity instance* consists of a finite variable set $\mathcal{X}$, a finite nonempty set $\mathcal{E}$ of equation occurrences, and positive rational weights ω_e summing to one. An occurrence is an ordered equation

$$x_{v_{e,1}} + x_{v_{e,2}} + x_{v_{e,3}} = b_e, \qquad b_e \in \mathbb{F}_2.$$

Its value is the weighted fraction of equations satisfied, and $\mathrm{OPT}(\mathcal{E})$ denotes the maximum value. Different occurrences have different IDs even if their ordered equations coincide.

Theorem 4.1 (Weighted parity gap)**.** *For every fixed rational $\xi > 0$, there is a deterministic polynomial-time reduction from Boolean satisfiability to weighted parity instances, with binary-encoded rational weights, such that*

$$\begin{aligned} \textit{YES:} \quad & \mathrm{OPT}(\mathcal{E}) \geq 1 - \xi, \\ \textit{NO:} \quad & \mathrm{OPT}(\mathcal{E}) \leq \tfrac{1}{2} + \xi. \end{aligned}$$

The running-time polynomial may depend on ξ.

This is the gap consequence of Håstad's construction in [18, Theorem 5.4 and its proof]. More specifically, the proof uses a fixed dyadic verifier error $\delta > 0$, gives completeness $1-\delta$ and soundness $(1+\delta)/2$, and enumerates verifier choices as equations with their rational probabilities as weights. Choosing $\delta \leq \xi$ gives the displayed statement. The enumeration is a deterministic reduction; the verifier's random choices describe the resulting weighted instance.

Lemma 4.2 (Making positions distinct)**.** *From a weighted parity instance $\mathcal{E}$ one can deterministically construct, in polynomial time, an instance $\mathcal{E}'$ in which the three variable IDs of every occurrence are distinct. Its weights are positive rationals of polynomial bit length, and*

$$\mathrm{OPT}(\mathcal{E}') \geq \mathrm{OPT}(\mathcal{E}), \qquad \mathrm{OPT}(\mathcal{E}') \leq \frac{1 + \mathrm{OPT}(\mathcal{E})}{2} + \frac{3}{100}.$$

In particular, Theorem 4.1 with $\xi < 1/100$, followed by this preprocessing, gives completeness at least $1 - \xi$ and NO optimum at most $4/5$.

Proof. Replace each variable v by one hundred clones (v, c), $c \in [100]$. For each original occurrence, choose its three clones uniformly subject to choosing distinct clones whenever positions have the same underlying variable. List all the resulting occurrences and give them the corresponding conditional weights. There are at most 100^3 choices per original occurrence. Giving all clones their original variable's label proves the completeness assertion.

For the upper bound fix arbitrary clone labels, and let

$$t_v = \frac{1}{100} \sum_{c=1}^{100} (-1)^{x_{(v,c)}}.$$

First choose clones independently, without conditioning on distinctness. The satisfaction probability of occurrence e is

$$\frac{1 + (-1)^{b_e} t_{v_{e,1}} t_{v_{e,2}} t_{v_{e,3}}}{2}.$$

Round each bias to a majority label, resolving zero biases arbitrarily. For every equation violated by this rounded assignment, the signed product in the numerator is nonpositive, so its satisfaction probability is at most $1/2$. The other equations have probability at most one. The weighted mean is therefore at most $(1+\mathrm{OPT}(\mathcal{E}))/2$. The probability of a forbidden collision among the independently chosen clones is at most $3/100$, by a union bound over the three pairs of positions. Conditioning on its complement changes the expectation of any $[0,1]$-valued function by at most $3/100$. This proves the bound. All conditional probabilities have constant-size denominators apart from their original occurrence weights. □

For the rest of the construction, $\mathcal{E}$ denotes the preprocessed instance, so every equation has three distinct variable IDs. Its *ordinary equation-versus-variable game* samples e according to ω and j uniformly from $[3]$. The first prover receives the occurrence ID e and answers with a satisfying

triple for that equation; the second receives the variable ID $v_{e,j}$ and answers with a bit. Acceptance means equality at position j. Because the IDs within an equation are distinct, the question pair determines the projection constraint.

For any second-prover strategy, its bits constitute a global assignment. On an equation violated by that assignment, every valid first-prover answer disagrees in at least one position. Hence this ordinary projection game, denoted $G_{\mathcal{E}}$, satisfies

$$\mathrm{val}(G_{\mathcal{E}}) \le 1 - \frac{1 - \mathrm{OPT}(\mathcal{E})}{3} \le \frac{14}{15} \quad \text{if } \mathrm{OPT}(\mathcal{E}) \le \frac{4}{5}. \tag{4.1}$$

The answer alphabets have sizes four and two, respectively.

4.2 Tuples, local answer copies, and homogeneous coordinates

Fix a positive integer k. An equation question $U = (e_1, \dots, e_k)$ is sampled from $\omega^{\otimes k}$, with replacement. Its answer space is the product of the k affine solution planes. Blocks always use local copies: even if the same equation or variable ID appears in several blocks, their local answers are separate coordinates.

For a parameter $0 < \beta < 1$, independently in every block either retain the equation, with probability $1 - \beta$, or retain just the variable ID at a uniformly chosen position, with probability β. The resulting question is denoted O, and π is the coordinate projection from answers on U to answers on O. The question O records an equation ID or a variable ID at every ordered block position. A singleton block accepts either bit.

Represent an equation answer by its first two bits (x_i, y_i); the third is $b_{e_i} + x_i + y_i$. A singleton is represented by its actual bit. Add a common homogeneous coordinate t_0. Thus

$$L_U = \mathbb{F}_2^{1+2k}, \qquad m = 1 + 2k,$$

and L_O has one homogeneous coordinate, two coordinates for each retained equation, and one for each singleton. In either space, vectors whose first coordinate is one are exactly valid answers. Write $e_0^\top$ for the functional reading that coordinate.

The affine projection π extends to a linear surjection $\bar{\pi} : L_U \to L_O$ preserving t_0. On retained equation blocks it is the identity, and on a block projected to a singleton its coordinate is

$$\begin{cases} x_i, & \text{position 1,} \\ y_i, & \text{position 2,} \\ b_{e_i} t_0 + x_i + y_i, & \text{position 3.} \end{cases} \tag{4.2}$$

We call every map obtained by these blockwise choices a *permitted projection*; this class of maps is independent of β. Surjectivity follows by fixing t_0 and choosing coordinates independently in each block. We use the same conventions for questions with some blocks omitted.

4.3 Reducing tables to their retained coordinates

The identification of tables across projected questions follows the virtual-table approach of Khot–Safra [28, Section 1.1]. Each table will be represented by its retained coordinates and the induced affine map. Using an equation's third bit as a coordinate requires adjusting the intercept. With that correction, a projected table and its pullback have the same representation.

Fix alphabet data $(\mathcal{V}, K, C, \mu)$ from Lemma 3.1, where $K = \mathbb{F}_2^\ell$, and fix a linear complement $\mathcal{V} = K \oplus W$. A *table* on a question Q is an affine map from its answer space to $\mathcal{V}$, or equivalently

its unique linear extension $P \in \mathrm{Hom}(L_Q, \mathcal{V})$. Here Q may be an equation question or one of its permitted projections, and the same definition applies to a question with omitted blocks. Write c for the coefficient of the homogeneous coordinate t_0 in P.

In an equation block, write the contribution of the slope map $\mathbb{F}_2^2 \to \mathcal{V}$ as $a_i x_i + d_i y_i$. Its rank determines which coordinates must be retained.

- If the rank is two, retain the equation occurrence and this contribution.
- If the rank is zero, omit the block.
- If the rank is one, the coefficients have exactly one of the forms

$$(a_i, d_i) = (v, 0), \quad (0, v), \quad (v, v), \qquad v \neq 0.$$

  These forms retain only the variable at position one, two, or three, respectively, with contribution vz_i in its actual bit z_i. For the third form,

$$v(x_i + y_i) = vz_i + b_{e_i} v t_0, \qquad z_i = b_{e_i} t_0 + x_i + y_i,$$

  so this replacement also adds $b_{e_i} v$ to the homogeneous intercept.

In a singleton block, retain its variable ID and its original contribution exactly when its slope is nonzero; otherwise omit the block.

The retained blocks form the *support* of (Q, P). They keep their indices in $[k]$, their equation/variable tags, and their IDs. A retained equation keeps its ordered first two coordinates, and a retained singleton uses its actual-bit coordinate. Thus repeated IDs in different blocks still have separate local coordinates. The support may be empty; its homogeneous coordinate remains in every case.

Let J_3 be the set of equation blocks reduced to their third bit, and write v_i for the nonzero slope in such a block. The *reduced table* $\widetilde{P}$ on the support has the retained block contributions specified above and homogeneous intercept

$$\widetilde{c} = c + \sum_{i \in J_3} b_{e_i} v_i. \tag{4.3}$$

The reduction preserves the table through a factorization. Let L_{supp} be the homogeneous answer space of the support and let $\rho_{Q,P} : L_Q \to L_{\mathrm{supp}}$ retain t_0 and the specified block coordinates. It is the identity on retained equations and on retained singleton blocks already present in Q, uses the corresponding actual-bit coordinate from (4.2) on a reduced equation, and discards omitted blocks. Choosing the coordinates independently in each block shows that $\rho_{Q,P}$ is surjective. The block identities and (4.3) give

$$P = \widetilde{P} \rho_{Q,P}.$$

Surjectivity makes $\widetilde{P}$ the unique linear map with this factorization. The *key* of (Q, P) is its support together with this entire reduced table. Thus two tables have the same key exactly when their retained coordinates agree, with the same positions, tags, and IDs, and their induced affine maps agree on those coordinates, including their intercepts.

Compatibility with projections. Let $\bar{\pi} : L_U \to L_O$ be any permitted projection and let $P' \in \mathrm{Hom}(L_O, \mathcal{V})$. An equation retained in O has the same contribution in P' and its pullback $P'\bar{\pi}$, so its reduction is identical. A singleton of slope zero disappears on both sides. A singleton of nonzero slope v pulls back to the corresponding one of the three rank-one patterns above, which reduces to

the same variable ID at the same block position. For a third-position singleton, the pullback adds $b_{e_i} v$ to the intercept, and its reduction adds $b_{e_i} v$ once more. The two terms cancel over $\mathbb{F}_2$. All other intercept corrections come from retained equation blocks and are identical on the two sides. The supports and complete reduced tables therefore agree: $(U, P'\bar{\pi})$ and (O, P') have the same key.

Define the key universe to consist of the keys of all tables on equation questions $U \in \mathcal{E}^k$. The compatibility just proved shows that the key of every table on a permitted projected question already belongs to this universe, by pullback from any compatible U.

4.4 Folding and shared table answers

The keys identify tables across projected questions. To obtain the vertices, we also identify keys that differ by a constant in K. The folding principle of Bellare, Goldreich, and Sudan [7, Section 3.3] then lets values on one representative determine the values on its translates. Adding a constant $h \in K$ to a key means adding h to its reduced intercept, leaving its support and slopes fixed. This is realized by replacing a presenting table P with $P + he_0^\top$, so it preserves the key universe. The action is free because the key records the intercept. A vertex of our permutation instance is an orbit of this action, represented by its unique key whose reduced intercept has zero K component.

For a table P on an equation question or a permitted projected question Q, let $c(Q, P) \in K$ be the K component of its key's reduced intercept. Let $v(Q, P)$ denote the orbit vertex represented by the key obtained after removing this component. A labeling λ of the vertices defines the restored table answer

$$F_Q(P) = \lambda(v(Q, P)) + c(Q, P).$$

Adding a constant changes only the removed component, so

$$F_Q(P + he_0^\top) = F_Q(P) + h \qquad (h \in K). \tag{4.4}$$

Tables with the same key have the same representative and removed component. The pullback compatibility therefore gives, for every permitted $U, O, \bar{\pi}$ and every table P' on O,

$$F_U(P'\bar{\pi}) = F_O(P'). \tag{4.5}$$

Thus a single vertex labeling determines answers on both the original and projected questions, with consistency supplied by the identity of their keys.

4.5 The matrix test and completeness

The permutation instance, denoted $\mathcal{G}$, samples independently

$$U \sim \omega^{\otimes k}, \quad P \text{ uniform in } \operatorname{Hom}(L_U, \mathcal{V}), \quad a \sim \mu, \quad l \text{ uniform in } \mathbb{F}_2^m,$$

and checks

$$F_U(P) = F_U(P + al^\top). \tag{4.6}$$

Here $al^\top$ denotes the map $z \mapsto a(l^\top z)$. Each sample specifies an edge between the two key-orbit vertices. If their restored offsets are c_0, c_1, its constraint is the permutation

$$\lambda(v_1) = \lambda(v_0) + c_0 + c_1$$

of the fixed alphabet K. Edge occurrences have their sampling probabilities as weights. At this stage loops and repeated edges are allowed. The sparse projection experiment involving O analyzes the same vertex labels; the edges of $\mathcal{G}$ are specified by (4.6).

Lemma 4.3 (Completeness of the outer matrix test)**.** *If $\mathcal{E}$ has an assignment of value at least $1-\xi$, and the nonlinear map satisfies*

$$\Pr_{x \text{ uniform in } \mathcal{V},\ a\sim\mu}[C(x+a) \neq C(x)] \leq p_*,$$

then

$$\mathrm{val}(\mathcal{G}) \geq 1 - k\xi - \frac{p_*}{2}.$$

Proof. Fix the near-satisfying global bit assignment. Give every reduced support a valid answer as follows. At a singleton use its global bit. At a retained equation use its first two global bits and set the third to the unique value satisfying the equation. These choices depend only on the support, including its occurrence IDs, and therefore define an answer for every key independently of its origin.

At an orbit representative, evaluate its reduced table on this answer and apply C. This defines a vertex labeling. The equivariance of C under K implies that its restored answer is evaluation of the actual reduced table followed by C, in agreement with (4.4).

With probability at least $1 - k\xi$, every equation in U is satisfied by the global assignment. On such a tuple, the support answers used for every table are projections of the same full answer $z_U \in L_U$, whose first coordinate is one. Consequently the two sides of (4.6) are

$$C(Pz_U) \quad \text{and} \quad C(Pz_U + a(l^\top z_U)).$$

Since $z_U \neq 0$, the vector Pz_U is uniform in $\mathcal{V}$, independently of a, l, and $l^\top z_U$ is an independent fair bit. The test therefore fails with probability at most $p_*/2$ on these tuples. Adding the probability of a tuple containing an unsatisfied equation proves the bound. □

5 Matrix extraction and Fourier decoding

We convert high acceptance of the matrix test into agreement between two strategies for the sparsely projected questions U, O. Each strategy uses only its own question and a bounded number of rows of matrix advice. The resulting agreement probability is positive independently of the tuple length k and the parity instance. The table functions enter only through their folding and sharing identities and the acceptance bound, so the proof applies to any family with these properties; the restored functions of every vertex labeling are examples. We state the precise assumptions below, after fixing the parameters and defining the advice experiment.

5.1 The inverse shortcode theorem

The structural input describes functions that often keep their value under a uniform rank-one perturbation. Such a function agrees with an affine evaluation on a set obtained by fixing a bounded number of linear combinations of rows and columns.

Write $\mathrm{Mat}_{\ell,m} = \mathrm{Hom}(\mathbb{F}_2^m, \mathbb{F}_2^\ell)$. A *row and column slice* is a nonempty set specified by equations

$$\mathcal{S} = \{M : d_i^\top M = s_i^\top \ (1 \leq i \leq n_r), \quad Mq_j = t_j \ (1 \leq j \leq n_c)\}, \tag{5.1}$$

where $d_i \in \mathbb{F}_2^\ell$, $s_i \in \mathbb{F}_2^m$, $q_j \in \mathbb{F}_2^m$, and $t_j \in \mathbb{F}_2^\ell$. The counts n_r, n_c need not be the ranks of their respective systems.

Theorem 5.1 (Equality soundness for degree-two shortcode)**.** *For every $\eta \in (0,1)$ there are $\alpha \in (0,1]$, an integer $r_s \geq 1$, and an integer ℓ_0, depending only on η, with the following property. For every $\ell \geq \ell_0$ there is $m_0 = m_0(\eta, \ell)$ such that, for every $m \geq m_0$ and every function $f : \mathrm{Mat}_{\ell,m} \to \mathbb{F}_2^\ell$, if*

$$\Pr_{M,a,l}[f(M) = f(M + al^\top)] \geq \eta, \tag{5.2}$$

then some slice $\mathcal{S}$ as in (5.1), with $n_r, n_c \leq r_s$, and some $z \in \mathbb{F}_2^m$, $u \in \mathbb{F}_2^\ell$ satisfy

$$\Pr_{M|\mathcal{S}}[f(M) = Mz + u] \geq \alpha. \tag{5.3}$$

Here M, $a \in \mathbb{F}_2^\ell$, and $l \in \mathbb{F}_2^m$ are independent and uniform, and $M \mid \mathcal{S}$ is uniform on the slice. No folding condition is imposed on f.

Proof. We pass from a frequently preserved fiber of f to a dense Grassmann interval, then express that interval as row and column constraints. We use the Grassmann expansion theorem of Khot–Minzer–Safra [25, Theorem 1.12 in the 2018 revision], together with the matrix chart of Barak–Kothari–Steurer [6, Definition 3.2 and Lemmas 3.3–3.7].

The Grassmann graph on an ambient binary space E has the ℓ-dimensional subspaces as vertices; two vertices are adjacent when their intersection has dimension $\ell - 1$. The retention of a nonempty vertex set is the probability that a uniform vertex in the set has a uniform neighbor still in the set. The cited theorem says that a set of retention at least a fixed $\zeta > 0$ has positive relative density in some interval

$$\{L : A_0 \subseteq L \subseteq B_0\}, \qquad \dim A_0 + \mathrm{codim}_E B_0 \leq r.$$

The density bound and r depend only on ζ. The theorem holds for all sufficiently large ℓ, and then all sufficiently large $\dim E$.

A frequently preserved fiber. Equality acceptance is the average of the retention probabilities of the nonempty fibers $f^{-1}(y)$ under the matrix step, weighted by their uniform masses. Some fiber S therefore has retention at least η. The probability that either perturbation factor is zero is

$$p_0 = 2^{-\ell} + 2^{-m} - 2^{-\ell-m}.$$

Choose the dimension thresholds so that $p_0 \leq \eta/2$. Conditional on both factors being nonzero, the retention of S is at least $(\eta - p_0)/(1 - p_0) \geq \eta/2$.

The Grassmann chart. For each matrix put

$$L_M = \{(x, M^\top x) : x \in \mathbb{F}_2^\ell\} \subseteq E := \mathbb{F}_2^\ell \oplus \mathbb{F}_2^m.$$

The map $M \mapsto L_M$ is a bijection onto the chart $\mathcal{C}$ of subspaces whose projection onto the first summand is invertible. Since

$$\dim(L_M \cap L_N) = \ell - \mathrm{rank}(M - N),$$

the neighbors within the chart correspond exactly to nonzero rank-one matrix differences. Each chart vertex has $(2^\ell - 1)(2^m - 1)$ such neighbors, whereas its full Grassmann degree is $(2^\ell - 1)(2^{m+1} - 2)$: choose a hyperplane in the vertex and then a one-dimensional extension in the quotient by that hyperplane, other than the vertex itself. Thus exactly half of its neighbors lie in the chart. Over $\mathbb{F}_2$, every nonzero rank-one matrix has a unique factorization $al^\top$ with nonzero factors. The lifted set $\widetilde{S} = \{L_M : M \in S\}$ consequently has Grassmann retention at least $\eta/4$.

From an interval to a slice. The expansion theorem with $\zeta = \eta/4$ gives density α on an interval with $\dim A_0 + \mathrm{codim}_E B_0 \leq r_s$. Intersecting this interval with $\mathcal{C}$ only increases the relative density

of $\widetilde{S}$, since $\widetilde{S} \subseteq \mathcal{C}$, and the intersection is nonempty. A basis (d_i, s_i) of A_0 specifies $A_0 \subseteq L_M$ by the equations $d_i^\top M = s_i^\top$. For the standard dot product on E, a basis (t_j, q_j) of $B_0^\perp$ specifies $L_M \subseteq B_0$ by $Mq_j = t_j$. Their simultaneous solution set is a slice of the required form, with at most r_s equations of either kind. On at least an α fraction of this slice, f equals the value defining S. This proves the conclusion with $z = 0$ and u equal to that value.

The constants α, r_s were chosen using only η. Since $\dim E = \ell + m$, after fixing ℓ the remaining dimension threshold is absorbed into $m_0(\eta, \ell)$. The fiber argument is the equality consequence in [6, Lemma 2.3]; numbered references to that paper use its arXiv version 1. □

The constants and dimension thresholds in Theorem 5.1 can be chosen effectively, with rational α when η is rational. This follows from the quantitative bound in [25, Theorem 2.13 and Section 2.4] and the comparison estimates in [25, Lemmas 2.7–2.8], with theorem numbers again referring to the 2018 revision.

The parameters are chosen in three stages. First choose $p_* > 0$ and the associated rank threshold r_* from Lemma 3.1. The inverse theorem then supplies α, r_s at the value of η below. Finally choose the dimension ℓ so that

$$\eta = 2^{-r_*}/16, \qquad 2^{-(\ell - r_s)} < \alpha/8, \qquad 2^{-\ell} < \eta/4, \tag{5.4}$$

and so that ℓ is sufficiently large for Theorem 5.1 and the alphabet lemma. At this dimension obtain the alphabet data $(\mathcal{V}, K, C, \mu)$, and fix a linear splitting $\mathcal{V} = K \oplus W$. All these objects are fixed before k grows. In particular, $\ell > r_s$. We write a table as $P = (M, T)$, with $M : L_U \to K$ and $T : L_U \to W$.

5.2 The advice experiment and the decoding statement

We first specify the local inputs for two strategies whose answer agreement will contradict the outer-game gap. This experiment is defined for every positive integer k and every $0 < \beta < 1$. Let $\mathcal{Q}_k$ consist of all equation questions $U \in \mathcal{E}^k$ and all permitted projected questions O obtained from them by the coordinate projections of Section 4. The questions U, O and the projection $\bar{\pi} : L_U \to L_O$ have the law in Section 4: independently in each block, an equation is retained with probability $1 - \beta$, and a uniformly chosen variable position is retained with probability β.

Definition 5.2 (Projected advice experiment)**.** *Sample U, O, π, then an independent uniform map $A : K \to \mathbb{F}_2^{r_s}$ and independent uniform maps*

$$T_O : L_O \to W, \qquad M_O : L_O \to K.$$

Put $S_O = AM_O$ and send the following inputs to the two provers:

$$\begin{aligned} &U\text{-prover}: \quad (U, A, T_O\bar{\pi}, S_O\bar{\pi}),\\ &O\text{-prover}: \quad (O, A, T_O, S_O). \end{aligned} \tag{5.5}$$

The map M_O beyond the stated advice is hidden. Each prover returns an actual answer, that is, a vector of first bit one in its homogeneous answer space. They succeed when their answers a_U, a_O satisfy $\bar{\pi} a_U = a_O$.

The U-message contains the maps $(T_U, S_U) = (T_O\bar{\pi}, S_O\bar{\pi})$, obtained by padding maps sampled on O. We will first find useful advice under a different law: an unrestricted uniform table (M, T) on U, with $S_U = AM$. Later we compare the full padded table $(M_O\bar{\pi}, T_O\bar{\pi})$ with this unrestricted table after averaging over the projection choices. This comparison is the bridge from the extraction argument to the experiment.

Proposition 5.3 (Matrix decoding)**.** *Choose the alphabet and shortcode parameters in the order specified by* (5.4)*, fixing* $\mathcal{V} = K \oplus W$ *before* k*. There are a constant* $\gamma > 0$ *and an integer* $k_0 \geq 8$*, depending only on these fixed parameters, with the following property.*

For every cube $k \geq k_0$ *and every parity instance, suppose a function*

$$F_Q : \mathrm{Hom}(L_Q, \mathcal{V}) \longrightarrow K$$

is given for every $Q \in \mathcal{Q}_k$*. Assume folding for every such* Q*, every* $P \in \mathrm{Hom}(L_Q, \mathcal{V})$*, and every* $c \in K$*, and assume sharing for every permitted projection* $\bar{\pi} : L_U \to L_O$ *and every* $P' \in \mathrm{Hom}(L_O, \mathcal{V})$*:*

$$\begin{aligned} F_Q(P + ce_0^\top) &= F_Q(P) + c, \\ F_U(P'\bar{\pi}) &= F_O(P'). \end{aligned} \tag{5.6}$$

If this family passes the table test of (4.6) *with probability at least* 0.99*, then the projected advice experiment with* $\beta = k^{-2/3}$ *has local strategies whose probability of full answer agreement is at least* γ*.*

Every labeling of the vertices of $\mathcal{G}$ supplies such a family: its restored functions satisfy (5.6) by (4.4) and (4.5). The two response rules may depend on the fixed table family. The proof first obtains good row advice and finite answer sets (Lemmas 5.5 and 5.6), transfers their slice guarantees to the projected experiment (Lemmas 5.7 and 5.8), and turns those guarantees into independent local agreement (Lemma 5.9). The deduction of Proposition 5.3 follows at the end of Section 5.7.

The next two subsections work for any positive integer k, with a fixed family satisfying (5.6). They use unrestricted uniform tables on U and do not yet restrict β. We specialize k and β when comparing that law with the projected experiment.

5.3 Comparing the two matrix perturbations

The uniform matrix perturbation has the rank-indexed Fourier spectrum computed in [25, Lemma 2.10 in the 2018 revision]. We compare that spectrum with the noise supplied by the alphabet construction.

Lemma 5.4 (Spectral comparison)**.** *For the parameters in* (5.4)*, if the table family passes* (4.6) *with probability at least* 0.99*, then*

$$\mathbb{E}_{U,T} \Pr_{M,a,l}[F_U(M,T) = F_U(M + al^\top, T)] \geq 0.9\, 2^{-r_*}, \tag{5.7}$$

where a *is uniform in* K *and the other matrix and vector samples are uniform in their respective spaces. Consequently at least* 10η *of the pairs* (U,T) *have the inner probability at least* 4η*.*

Proof. High acceptance forces most Fourier mass onto frequencies whose restrictions to K have small rank. Such frequencies also have a positive eigenvalue for a uniform perturbation in K.

For $\sigma \in K^*$ define the sign function $f_{U,\sigma}(P) = (-1)^{\sigma(F_U(P))}$. A frequency on the matrix domain is a tuple $g = (g_i)_{i=0}^{m-1}$ of functionals in $\mathcal{V}^*$; its character at P is $(-1)^{\sum_i g_i(Pe_i)}$. We use the normalized Fourier coefficients of Section 2. Translation by $al^\top$ multiplies this character by $(-1)^{\sum_i l_i g_i(a)}$. Averaging over the uniform selector l gives the two Fourier multipliers

$$\begin{aligned} \mathbb{E}_{a\sim\mu,l}(-1)^{\sum_i l_i g_i(a)} &= \Pr_{a\sim\mu}[g_i(a) = 0 \text{ for every } i], \\ \mathbb{E}_{a \text{ uniform in } K,l}(-1)^{\sum_i l_i g_i(a)} &= 2^{-\dim \mathrm{span}\{g_i|_K\}}. \end{aligned}$$

Write the first multiplier as

$$\Lambda_g = \Pr_{a\sim\mu}[g_i(a) = 0 \text{ for every } i].$$

It lies in $[0, 1]$. By Lemma 3.1, it is at most $7/8$ whenever

$$\dim \operatorname{span}\{g_i|_K : 0 \le i < m\} \ge r_*.$$

Indeed the span of the g_i is a subspace of $\mathcal{V}^*$ to which the lemma applies.

Expanding equality by a uniform output character, and then using Fourier orthogonality, expresses the test acceptance as

$$\mathbb{E}_{U,\sigma} \sum_g |\widehat{f}_{U,\sigma}(g)|^2 \Lambda_g.$$

Parseval gives total squared mass one for each sign function. If B is the average squared mass on restricted ranks at least r_*, then $0.99 \le 1 - B/8$, so $B \le 0.08$. In particular, at least 0.9 of the average mass lies on ranks below r_*.

When a is instead uniform in K, the eigenvalue at g is $2^{-\dim \operatorname{span}\{g_i|_K\}}$. The preceding mass thus gives the lower bound in (5.7). This perturbation leaves T fixed, so the expression there is precisely the average of the ordinary shortcode equality tests at fixed (U, T). Its lower bound is 14.4η. If q is the probability that the inner acceptance is at least 4η, then the average is at most $4\eta + q$. Thus $q \ge 10.4\eta \ge 10\eta$. □

5.4 Row fibers with agreeing slices

The inverse theorem gives one slice on each qualifying (U, T) fiber where the table agrees with an affine target. The U-prover's information about the K-valued matrix is the pair (A, AM). We need a positive probability that these observed rows determine a useful slice. We will also identify the short set of valid answers supplied by that slice. An erasure argument proves the probability bound in Lemma 5.5; folding supplies the nonempty answer set in Lemma 5.6.

For this extraction step, sample U as before and take independent uniform maps $T : L_U \to W$, $A : K \to \mathbb{F}_2^{r_s}$, and $M : L_U \to K$. For fixed U, T, A and a map $S_0 : L_U \to \mathbb{F}_2^{r_s}$, the set $\{M : AM = S_0\}$ is its *row fiber*. We call the row advice (U, T, A, S_0) good if there exist at most r_s column specifications and a target $Mz + u$ such that the simultaneous solution set of the column specifications and $AM = S_0$ is nonempty and $F_U(M, T) = Mz + u$ with probability at least $\alpha/2$ under the uniform measure on this set.

Goodness imposes an agreement condition on the row fiber; it initially allows any target vector z, including vectors that are not actual answers. The fixed column values allow z to vary modulo the span of their column directions, with a corresponding change in u. Thus the slice naturally supplies a coset of possible target vectors. Its valid part will be the answer set sampled by the U-prover.

Lemma 5.5 (Row advice with an agreeing slice)**.** *For the parameters in* (5.4) *and all sufficiently large k, a table family satisfying* (5.6) *with matrix-test acceptance at least* 0.99 *satisfies*

$$\Pr_{U,T,A,M}[(U, T, A, AM) \text{ is good}] \ge g_0, \qquad g_0 = 10\eta^2\, 2^{-\ell r_s} > 0. \tag{5.8}$$

The lower bound and the required largeness of k are independent of the parity instance and the family.

Proof. We show that good row fibers cover a positive fraction of the matrix domain. If their union were too small, randomizing it would preserve high test acceptance while reducing agreement below α on every allowed slice, contradicting Theorem 5.1.

Randomizing the good fibers. Fix (U, T) for which the uniform rank-one acceptance is at least 4η, and write $f(M) = F_U(M, T)$. Let B be the union, in the matrix domain, of all good row fibers, over every possible A, S_0. Suppose that its uniform measure is less than η. Change f at the entries of B, replacing each by an independent uniform element of K. Each matrix receives one replacement, even when several good row fibers overlap there. Both endpoints of the shortcode step are uniform matrices. Consequently every such modification changes acceptance by at most $2\Pr[B]$, and the modified function has acceptance at least 2η.

Controlling all slice targets. The random replacements can be chosen so that they contribute less than $\alpha/4$ of the size of every simultaneous row and column slice as matches with every target $Mz + u$. A nonempty slice with at most r_s row and r_s column specifications has homogeneous space

$$\operatorname{Hom}\Big(\mathbb{F}_2^m / \operatorname{span}\{q_j\}, \bigcap_i \ker d_i^\top\Big).$$

Its size $n_{\mathcal{S}}$ is therefore at least $2^{(\ell - r_s)(m - r_s)}$. Padding the lists of specifications with zero equations, the number of slice–target descriptions is at most $2^{(2r_s+1)(\ell+m)}$.

For a fixed description, count only matches at randomized entries and assign a deterministic zero contribution to the other entries. These are independent variables in $[0, 1]$, with total mean at most $2^{-\ell} n_{\mathcal{S}} < \alpha n_{\mathcal{S}}/8$. Hoeffding's inequality [19, Theorem 2, equation (2.6)] bounds the probability of at least $\alpha n_{\mathcal{S}}/4$ such matches by $\exp(-\alpha^2 n_{\mathcal{S}}/32)$. The union bound over all descriptions is thus at most

$$\exp\left((2r_s + 1)(\ell + m)\log 2 - \frac{\alpha^2}{32} 2^{(\ell - r_s)(m - r_s)}\right),$$

which tends to zero as $m = 1 + 2k$ grows. Fix replacements with the asserted property.

Contradiction and averaging. Consider any slice of the allowed type. If its row fiber was good, every entry of that fiber belongs to B, and its modified agreement is less than $\alpha/4$. If the row fiber was not good, the original agreement with each target on the slice was less than $\alpha/2$. The unmodified matches form a subset of those original matches, and the randomized matches add less than $\alpha/4$. The modified agreement is then less than $3\alpha/4$. Padding shorter row systems by zeros shows that these two cases cover every slice in Theorem 5.1. That theorem now contradicts the modified function's acceptance at least 2η. Theorem 5.1 applies to the modified function because it permits arbitrary functions, including those that fail folding.

It follows that $\Pr[B] \geq \eta$. For each $M \in B$, at least one matrix A has a good row fiber containing M. The uniform choice of A hits that particular matrix with probability $2^{-\ell r_s}$. Hence, at this fixed (U, T), the probability of good data is at least $\eta 2^{-\ell r_s}$. Lemma 5.4 supplies at least 10η of such pairs (U, T), proving (5.8). All counting and concentration bounds depended only on the fixed parameters and m, as required. □

Lemma 5.6 (A finite set of valid answers)**.** *Fix the parameters in* (5.4)*. For every positive integer k and every family satisfying* (5.6)*, one can select deterministically for each good row advice at most r_s column specifications $Mq_j = t_j$, with $q_j \in L_U$ and $t_j \in K$, and a target $Mz + u$, with $z \in L_U$, $u \in K$, and $e_0^\top z = 1$, satisfying the definition of goodness. Writing $Z = \operatorname{span}\{q_j\}$, the chosen witness determines the answer set*

$$\mathcal{A} = (z + Z) \cap \{v \in L_U : e_0^\top v = 1\}, \qquad 1 \leq |\mathcal{A}| \leq 2^{r_s}. \tag{5.9}$$

The witness and $\mathcal{A}$ depend only on (U, T, A, S_0) and the fixed table family. The slice and its target retain the agreement guarantee in the definition of goodness.

Proof. Fix any positive integer k, a family satisfying (5.6), and any good row advice. Fix a witness, write its column specifications as $Mq_j = t_j$, and put $Z = \operatorname{span}\{q_j\}$. Nonemptiness of the slice makes the prescribed values consistent: they determine a linear map $t : Z \to K$ with $t(w) = Mw$ throughout the slice. Replacing z by $z + w$ and u by $u + t(w)$ for $w \in Z$ therefore preserves the target. It suffices to show that the coset $z + Z$ contains a vector of first bit one.

Otherwise z and every vector in Z have first bit zero. Put $H = \ker A$. For each $h \in H$, translation $M \mapsto M + he_0^\top$ preserves the row fiber, the column specifications, and the target. By folding it translates $F_U(M, T)$ by h. The action is free, and in each of its H-orbits at most one matrix can satisfy the target equality. The agreement is at most

$$|H|^{-1} = 2^{-\ell + \operatorname{rank} A} \le 2^{-(\ell - r_s)} < \alpha/8,$$

contradicting goodness. Choose $w \in Z$ for which $z + w$ has first bit one and make the corresponding changes to z and u. They preserve the original agreement fraction. Choose a witness and such a representative using a fixed order on the finite possibilities. This choice uses only the row advice and the fixed table family. Its vector z lies in $\mathcal{A}$, so the set is nonempty; since $\dim Z \le r_s$, its size is at most 2^{r_s}. □

5.5 Transferring the chosen slice

The comparison with sparsely projected questions follows the smooth outer-game method of Khot–Safra [28, Section 3 and Lemma 3.1], with shared linear advice as in Khot–Minzer–Safra [26, Sections 3.3–3.4]. Lemma 5.5 gives a positive probability of good row advice under unrestricted sampling on U, and Lemma 5.6 assigns a short set of valid answers to each such datum. In the projected experiment, the maps are sampled on O and padded to U. We compare the complete padded table with the unrestricted table, including the affine intercept, with projection choices averaged out, and then transfer the chosen slice to O.

From now on take k along integer cubes larger than one and set $\beta = k^{-2/3}$ in Definition 5.2.

The choice $\beta = k^{-2/3}$ serves two purposes: $k\beta^2 \to 0$ makes the padded table law approach the unrestricted law in Lemma 5.7, while $k\beta \to \infty$ makes the expected number of singleton coordinates diverge. Once the advice spaces are fixed, each singleton has a uniformly positive probability that every visible advice row has zero coefficient on its variable bit. Lemma 6.2 uses these coordinates to bound agreement.

Lemma 5.7 (The padded matrix is asymptotically uniform)**.** *In Definition 5.2 with $\beta = k^{-2/3}$, conditional on every U, the distribution of*

$$(M, T) = (M_O\bar{\pi}, T_O\bar{\pi})$$

has total variation distance $o(1)$ from the unrestricted uniform distribution on $\operatorname{Hom}(L_U, K) \times \operatorname{Hom}(L_U, W)$, as k tends to infinity along cubes. The bound is uniform in U and the parity instance. The projection choices are averaged out in this comparison.

Proof. Combine the two components into a $\mathcal{V}$-valued table and put $n = |\mathcal{V}|$. On an unprojected equation block the two slopes are independent uniform elements of $\mathcal{V}$. On a singleton block the padded pair of slopes is an equal mixture of

$$(a, 0), \qquad (0, a), \qquad (a, a), \qquad a \text{ uniform in } \mathcal{V}.$$

Call these pair laws P_0 and Q_0, respectively. The likelihood ratio dQ_0/dP_0 is n at $(0,0)$, $n/3$ at the other $3(n-1)$ points on the three displayed lines, and zero elsewhere. Thus

$$\chi^2(Q_0\|P_0) := \mathbb{E}_{P_0}\left[\left(\frac{dQ_0}{dP_0}-1\right)^2\right] = \frac{n-1}{3}.$$

The one-block mixture $(1-\beta)P_0+\beta Q_0$ consequently has chi-squared divergence $\beta^2(n-1)/3$ from P_0. Blocks are sampled independently, including when their question IDs coincide. Multiplying their likelihood ratios gives divergence

$$\left(1+\frac{n-1}{3}\beta^2\right)^k - 1 = o(1), \tag{5.10}$$

because n is fixed and $k\beta^2 = k^{-1/3}$ tends to zero. Total variation is at most one half the square root of this expression.

For a projection onto the third bit of equation e, the padded intercept acquires the term $b_e a$, where a is that singleton's slope. The original intercept is independent and uniform in $\mathcal{V}$. Adding the sum of these terms leaves it uniform and independent even after all projection choices and slopes are given. The preceding slope estimate therefore proves the assertion for the entire table. The slope laws and the bound in (5.10) are uniform over the equation IDs and their right-hand sides. □

Fix good U-advice for a moment, with its chosen witness and answer set $\mathcal{A}$ from Lemma 5.6. For a permitted projection put $z' = \bar{\pi}z$ and $Z' = \bar{\pi}Z$. Since $\bar{\pi}$ preserves the first coordinate,

$$\bar{\pi}\mathcal{A} = (z'+Z') \cap \{v \in L_O : e_0^\top v = 1\}. \tag{5.11}$$

Indeed, every vector in the coset on the right has a preimage in $z+Z$, and the preimage has the same first coordinate. We will show that the O-prover's private Fourier sample reaches this projected answer set.

The certifying slice does not automatically survive projection. For example, a column condition $Mq = t$ becomes $M_O\bar{\pi}q = t$; it is inconsistent if $\bar{\pi}q = 0$ but $t \neq 0$. More generally, projected column directions may acquire new dependencies. The next argument uses the comparison of the full table laws to find a positive probability of samples where the chosen slice remains nonempty and retains enough agreement.

For this analysis, write $D = (U, O, \pi, A, T_O, S_O)$ for the joint data that determine both prover messages and the projection. Conditional on these data, M_O remains uniform on the row fiber $AM_O = S_O$.

Lemma 5.8 (Transfer of a chosen witness)**.** *For the parameters in* (5.4)*, suppose the table family satisfies* (5.6) *and has matrix-test acceptance at least* 0.99*. For all sufficiently large cubes k, a set of probability at least*

$$\gamma_1 = \frac{\alpha g_0}{8}2^{-\ell r_s} > 0 \tag{5.12}$$

of the joint data D in Definition 5.2 has the following properties. The padded U-advice is good, with the chosen answer set $\mathcal{A}$ and its certifying witness. Writing $z' = \bar{\pi}z$, that witness transfers to a nonempty slice

$$AM_O = S_O, \qquad M_O\bar{\pi}q_j = t_j,$$

on which

$$\Pr[F_O(M_O, T_O) = M_O z' + u \mid AM_O = S_O,\ M_O\bar{\pi}q_j = t_j \text{ for all } j] \geq \alpha/4. \tag{5.13}$$

The probability in this equation is taken over a uniform matrix in the specified slice.

Proof. We compare the difference of two event probabilities under the two table laws, then condition on the joint messages to locate the transferred slices.

Comparison before conditioning. Under unrestricted sampling of U, A, M, T, put $S_0 = AM$. Let $\mathcal{J}$ be the event that (U, T, A, S_0) is good and M satisfies its chosen column specifications. Let $\mathcal{M}$ be equality with the chosen target, declaring this event false off good data. Within a good row fiber the chosen nonempty column slice from Lemma 5.6 imposes at most ℓr_s additional scalar linear equations. Its agreement guarantee and the probability bound in Lemma 5.5 give

$$\Pr[\mathcal{J}] \geq g_0 2^{-\ell r_s}, \qquad \Pr[\mathcal{J} \cap \mathcal{M}] \geq (\alpha/2) \Pr[\mathcal{J}].$$

It follows that

$$\Pr[\mathcal{J} \cap \mathcal{M}] - (\alpha/4) \Pr[\mathcal{J}] \geq \frac{\alpha g_0}{4} 2^{-\ell r_s}. \tag{5.14}$$

Both events in this expression depend only on (U, A, M, T). Adding the independent A to Lemma 5.7 preserves its variation bound. We may therefore compare the two event probabilities with their values in the padded experiment. If the total variation distance is $v_k = o(1)$, the difference in (5.14) changes by at most $(1+\alpha/4)v_k$. For all sufficiently large k the padded difference is at least γ_1.

Locating the transferred slices. Conditional on D, the remaining M_O is uniform on the row fiber $AM_O = S_O$. The padded U-advice, and therefore its chosen witness and answer set, are fixed. On good data let p_D be the conditional probability of the projected column slice, and, when $p_D > 0$, let q_D be its conditional equality fraction. Put $p_D = 0$ on data that are not good, and put $q_D = 0$ when $p_D = 0$. Sharing gives $F_U(M_O\bar{\pi}, T_O\bar{\pi}) = F_O(M_O, T_O)$, so the padded difference is exactly

$$\mathbb{E}_D[p_D(q_D - \alpha/4)] \geq \gamma_1.$$

The integrand is nonpositive outside the event consisting of goodness, $p_D > 0$, and $q_D \geq \alpha/4$, and it is at most one everywhere. That event thus has probability at least γ_1 and has precisely the asserted properties. The law comparison applied to events depending only on (U, A, M, T); conditioning on D here only identifies where their positive difference occurs in the projected experiment. □

5.6 The two local decoders

Extracting an outer strategy from Fourier mass follows the approach of Håstad [18, Section 1.1] and the Fourier-list method of Khot–Safra [28, Section 1.1]. Here the projected prover samples squared Fourier mass on its own row fiber. The transferred slice will show that this private sample reaches the projected answer set $\bar{\pi}\mathcal{A}$ with positive probability.

The U-prover uses its visible row advice to make the deterministic choice in Lemma 5.6. When the advice is good, it samples a_U uniformly from $\mathcal{A}$. This is a nonempty set of at most 2^{r_s} actual answers. On other inputs the prover returns any actual answer.

The O-prover starts from its own row fiber. Given (O, A, T_O, S_O), put $H = \ker A$ and choose, by a fixed rule, a representative M_0 satisfying $AM_0 = S_O$. Such a representative exists on every input in the experiment. Write the row fiber as $M_0 + X$, where $X = \mathrm{Hom}(L_O, H)$. Sample uniform $\sigma \in K^*$ and form the sign function

$$f_\sigma(N) = (-1)^{\sigma(F_O(M_0+N, T_O))}, \qquad N \in X.$$

Sample a Fourier frequency $\Phi \in X^*$ with probability $|\widehat{f_\sigma}(\Phi)|^2$. Identify $X^* = H^* \otimes L_O$, with the pairing defined on pure tensors by

$$\langle \varphi \otimes v, N \rangle = \varphi(Nv).$$

For $h \in H$, contraction sends Φ to a vector $\Phi(h) \in L_O$. If $\sigma|_H \neq 0$, choose by a fixed rule an $h \in H$ with $\sigma(h) = 1$ and return $\Phi(h)$. If the restriction is trivial, return any actual answer.

This rule always returns an actual answer at frequencies of positive mass. Indeed the folding identity in (5.6) gives

$$f_\sigma(N + he_0^\top) = -f_\sigma(N) \qquad \text{when } \sigma(h) = 1.$$

Changing variables in a Fourier coefficient shows that its being nonzero requires $(-1)^{\langle \Phi, he_0^\top \rangle} = -1$, equivalently $e_0^\top \Phi(h) = 1$.

Figure 3 summarizes the information and private choices of the two decoders.

5.7 Agreement from Fourier mass

The transferred slice yields correlation on an affine restriction of the projected row fiber. Fourier expansion turns this correlation into a lower bound on the mass sampled by the projected decoder.

Lemma 5.9 (Agreement from a transferred slice)**.** *Fix joint data $D = (U, O, \pi, A, T_O, S_O)$ with the properties of Lemma 5.8. Conditional on these data, the two decoders of Section 5.6, using independent randomness, agree under $\bar{\pi}$ with probability at least*

$$2^{-\ell r_s}(\alpha/8)^2 2^{-r_s}. \tag{5.15}$$

Proof. Correlation on the slice. Recall $Z' = \bar{\pi}Z$, so $\dim Z' \leq r_s$. The projected column specifications in the row fiber fix the restriction $N|_{Z'}$ to a linear map $b : Z' \to H$. The resulting affine slice is nonempty; choose one N_* in it. For every $\sigma \in K^*$ define

$$C_\sigma = \mathbb{E}\big[f_\sigma(N)(-1)^{\sigma(Nz')} \mid N|_{Z'} = b\big].$$

The character expansion of (5.13) gives

$$\mathbb{E}_\sigma\big[(-1)^{\sigma(M_0 z' + u)} C_\sigma\big] \geq \alpha/4, \qquad \text{so} \qquad \mathbb{E}_\sigma|C_\sigma| \geq \alpha/4.$$

Exactly a fraction $2^{-\dim H}$ of the characters have trivial restriction to H. Since $\dim H \geq \ell - r_s$ and $|C_\sigma| \leq 1$, (5.4) implies

$$\mathbb{E}_\sigma\big[\mathbf{1}_{\{\sigma|_H \neq 0\}} |C_\sigma|\big] \geq \alpha/8. \tag{5.16}$$

Fourier mass on a coset. The annihilator in X^* of matrices vanishing on Z' is $H^* \otimes Z'$. For $\theta_\sigma = (\sigma|_H) \otimes z'$, Fourier expansion on the affine slice therefore gives

$$C_\sigma = \sum_{\Phi \in \theta_\sigma + H^* \otimes Z'} \widehat{f}_\sigma(\Phi)(-1)^{\langle \Phi + \theta_\sigma, N_* \rangle}.$$

This sum has $2^{\dim H \dim Z'} \leq 2^{\ell r_s}$ terms. Cauchy–Schwarz, first on each sum and then in the average over σ, shows from (5.16) that

$$\begin{aligned} \mathbb{E}_\sigma \left[\mathbf{1}_{\{\sigma|_H \neq 0\}} \sum_{\Phi \in \theta_\sigma + H^* \otimes Z'} |\widehat{f}_\sigma(\Phi)|^2 \right] &\geq 2^{-\ell r_s} \mathbb{E}_\sigma\big[\mathbf{1}_{\{\sigma|_H \neq 0\}} C_\sigma^2\big] \\ &\geq 2^{-\ell r_s}(\alpha/8)^2. \end{aligned}$$

This is a lower bound on the probability that the O-decoder samples a nontrivial restriction and a frequency in the indicated coset.

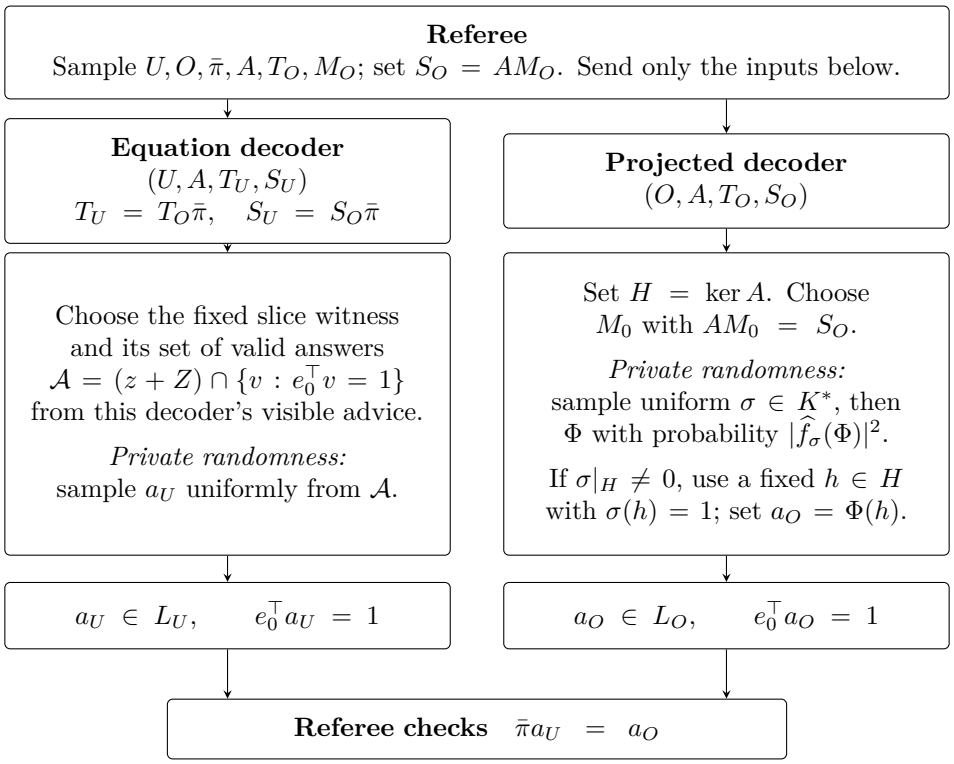


Figure 3: The two decoders use independent random tapes. Neither receives the sampled M_O or the projection $\bar{\pi}$; the right decoder chooses its own representative M_0 from the visible advice. Every displayed advice map includes its affine intercept, through the homogeneous coordinate. The transferred slice is used only to analyze these private rules: the right decoder does not use the left witness or answer set. If the left advice is not good, or if $\sigma|_H = 0$ on the right, that decoder returns an arbitrary valid answer.

Agreement of the answers. Every such frequency has the form $\Phi = (\sigma|_H) \otimes z' + \Psi$ with $\Psi \in H^* \otimes Z'$. For the decoder's choice of h it satisfies

$$\Phi(h) = z' + \Psi(h) \in z' + Z'.$$

The folding calculation in Section 5.6 shows that a frequency of positive mass gives first bit one. By (5.11), the returned vector lies in $\bar{\pi}\mathcal{A}$. Independently, the U-decoder samples uniformly from $\mathcal{A}$ and chooses a preimage with probability at least $1/|\mathcal{A}| \geq 2^{-r_s}$. Multiplying the bounds proves (5.15). □

Proof of Proposition 5.3. For the two decoders of Section 5.6, Lemma 5.8 supplies a set of joint data of probability at least γ_1, and Lemma 5.9 gives the stated conditional agreement on that set. Thus one may take

$$\gamma = \gamma_1 2^{-\ell r_s} (\alpha/8)^2 2^{-r_s} > 0.$$

The largeness requirements for k come only from Theorem 5.1, the concentration bound in Lemma 5.5, and the total variation estimate in Lemma 5.7. They are uniform in the parity instance and table family. Witness choices and Fourier sampling need not be efficient: they define strategies in a finite game for the soundness argument. Finally, averaging over the provers' independent random tapes also gives deterministic strategies with at least the same agreement probability.

□

6 Vanishing advice and the soundness gap

Proposition 5.3 converts high acceptance of the matrix test into agreement between two local strategies supplied with correlated linear advice. On a NO parity instance, the value of this advice experiment tends to zero. The use of vanishing advice is part of the smooth outer-game framework of [26, Sections 3.3–3.4]. For the present occurrence-indexed questions, the key is to identify singleton coordinates on which both advice slopes vanish. After fixing the data outside those coordinates, their question pairs have the ordinary independent equation–variable law. Each side can then reconstruct its original advice input from its own remaining questions and the fixed data. This defines strategies for an ordinary parallel repetition, whose value bounds the original agreement.

6.1 Classical projection-game repetition

Recall the classical game conventions of Section 2. We use the following alphabet-independent form of the parallel repetition theorem of Dinur and Steurer.

Theorem 6.1 (Projection-game repetition)**.** *Let G be a finite bipartite projection game, with an arbitrary probability distribution on its constraint occurrences. Question sets and answer alphabets may be arbitrary finite sets, and parallel constraint occurrences are allowed. The game $G^{\otimes n}$ samples n independent occurrences, sends each prover its tuple of questions, and accepts if all n projection constraints hold. For every integer $n \geq 1$ and every $0 < g < 1$,*

$$\mathrm{val}(G) \leq 1 - g \quad \Longrightarrow \quad \mathrm{val}(G^{\otimes n}) \leq (1 - g^2/16)^n. \tag{6.1}$$

The bound is independent of the question-set and alphabet sizes.

This is Corollary 1.2 of [13]. Their Section 2.1 permits nonnegative edge weights and parallel edges, and Section 2.2 defines the independent product used here. Thus the theorem applies to any finite weighted question distribution in our constructions. Interchanging the provers, if necessary, matches the direction of the projection.

6.2 Coordinates with vanishing advice

Let $\mathcal{E}$ be the parity instance from Section 4, with three distinct variable IDs in each equation occurrence. Recall its ordinary equation–variable game $G_{\mathcal{E}}$: the first prover receives an equation and answers with a satisfying triple; the second receives a uniformly selected variable from that equation and answers with a bit. On a NO instance, (4.1) gives $\mathrm{val}(G_{\mathcal{E}}) \le 14/15$.

Recall Definition 5.2, with k an integer cube larger than one and $\beta = k^{-2/3}$. The questions U, O and projection $\bar{\pi}$ have the independent blockwise law from Section 4. The advice is generated from the independent uniform maps

$$A : K \to \mathbb{F}_2^{r_s}, \qquad T_O : L_O \to W, \qquad M_O : L_O \to K$$

by putting $S_O = AM_O$. Recall that the two inputs are

$$(U, A, T_O\bar{\pi}, S_O\bar{\pi}) \quad \text{and} \quad (O, A, T_O, S_O). \tag{6.2}$$

The matrix M_O beyond S_O is not part of either input. Each prover must return an actual answer, equivalently a vector whose first coordinate is one in its respective answer space.

Lemma 6.2 (Agreement bound for the advice experiment)**.** *Let $k > 1$ be an integer cube and put $\beta = k^{-2/3}$. If $\mathrm{OPT}(\mathcal{E}) \le 4/5$, every pair of local randomized strategies in the experiment of Definition 5.2 satisfies*

$$\Pr[\bar{\pi}(a_U) = a_O] \le \left(1 - \frac{\beta\, 2^{-\dim W - r_s}}{3600}\right)^k \le \exp\left(-\frac{2^{-\dim W - r_s}}{3600} k^{1/3}\right). \tag{6.3}$$

In particular, the upper bound tends to zero uniformly over the parity instance as k tends to infinity along cubes, with the advice spaces fixed.

Proof. Fix an arbitrary pair of local randomized strategies. We fix the data outside the singleton coordinates with zero advice, reconstruct both original local inputs from the remaining questions, and apply parallel repetition to those questions.

The coefficient law. Conditional on A, combine the two visible advice maps into

$$D_O = (T_O, S_O) : L_O \longrightarrow E_A, \qquad E_A = W \oplus \operatorname{im} A.$$

Conditional on the questions and projection choices, the intercept c_O and all block slope columns of D_O are independent uniform elements of E_A. Indeed, the columns of T_O and M_O are independent and uniform, and applying A to a uniform column gives a uniform element of $\operatorname{im} A$. This coefficient law depends only on whether a block is an equation or a singleton, and not on any equation ID, variable ID, or selected position. In particular, c_O is independent of all block data.

Call coordinate i *clean* if it is projected to a singleton and its one advice slope $d_i \in E_A$ is zero. Conditional on A, the clean indicators are independent, with common probability

$$p_A = \frac{\beta}{|E_A|} = \beta\, 2^{-\dim W - \operatorname{rank} A}. \tag{6.4}$$

More precisely, for every equation occurrence e and position $j \in [3]$,

$$\Pr[e_i = e,\ i \text{ is a clean singleton at position } j \mid A] = \omega_e \frac{\beta}{3|E_A|}. \tag{6.5}$$

Thus, conditional on being clean, a coordinate has precisely the ordinary equation–variable distribution.

Conditioning outside the clean set. Let $I \subseteq [k]$ be the clean set. For the analysis, let R consist of A, I, the original intercept c_O, and the following data for every $i \notin I$: its equation occurrence, whether the block was retained or projected, the selected position in the latter case, and its advice slope columns. No witness, decoder output, agreement event, or response randomness is included in R.

For each value $R = r$ of positive probability, regard the clean set I recorded in r as fixed. When I is nonempty, we will use r as constant data when defining two strategies for $G_{\mathcal{E}}^{\otimes|I|}$. Each strategy will reconstruct one original advice input from its own product-game questions, apply the corresponding original response rule, and retain the answers in the clean slots.

Conditional on $R = r$, the pairs

$$(e_i, v_{e_i,j_i}), \qquad i \in I,$$

are independent question pairs of $G_{\mathcal{E}}$. Indeed, conditional on A, the coordinate data are independent, and (6.5) shows that conditioning on the clean mask preserves the ordinary law in each clean coordinate. The outside coordinate data and the independent intercept impose no further condition on these pairs.

Reconstructing the local inputs. The reconstruction must account for the affine intercept shared across the blocks. An equation with right-hand side b_e has actual bits $x, y, b_e t_0 + x + y$. Pulling back a singleton slope d_i thus produces the two slopes

$$(d_i, 0), \quad (0, d_i), \quad (d_i, d_i) \tag{6.6}$$

at positions one, two, and three, respectively. The intercept of $D_U = D_O \bar{\pi}$ is

$$c_U = c_O + \sum_{\substack{i \text{ projected to a singleton} \\ j_i = 3}} b_{e_i} d_i. \tag{6.7}$$

For a clean coordinate, every slope in (6.6) is zero and its contribution to (6.7) is zero. All remaining terms in that equation are determined by r.

The equation side receives its occurrences e_i for $i \in I$. Together with r, they specify all of U. Its two advice slopes in each clean block are zero, while r determines every other slope and the full intercept. These coefficients reconstruct both padded advice maps without using any clean selected position or variable question.

The variable side receives the variable question for each $i \in I$. Together with r, these specify all of O. Its clean singleton slopes are zero, and r supplies every other slope and the original intercept. These coefficients reconstruct its two advice maps without using any clean equation. Repeated IDs across different blocks cause no additional restriction: questions were sampled independently with replacement, and answers use the local copies specified in Section 4.

Repetition and averaging. These reconstructions define the product-game strategies: each side applies its original response rule to the reconstructed input and retains its answers in the slots I, in their original order. Their equation answers are satisfying triples because the original answers are valid. For every realization of the clean questions and response randomness, full agreement in the advice experiment implies acceptance in every clean slot. If $I = \varnothing$, the conditional success probability is at most one. Otherwise,

$$\Pr[\bar{\pi} a_U = a_O \mid R = r] \le \mathrm{val}(G_{\mathcal{E}}^{\otimes|I|}) \le \left(1 - \frac{1}{3600}\right)^{|I|}.$$

The first inequality uses the strategies just constructed and the conditional product law. The second uses Theorem 6.1 with the gap $g = 1/15$ from (4.1), so $g^2/16 = 1/3600$. The original response rules are evaluated only at their original local inputs; r indexes the auxiliary product-game strategies. Thus the final bound holds for every r, including $I = \varnothing$ when its right-hand side is interpreted as one.

Finally, conditional on A, the size of I has distribution Binomial(k, p_A). Averaging the preceding inequality first over R conditional on A, and then over A, gives

$$\begin{aligned}\Pr[\bar{\pi}(a_U) = a_O] &\le \mathbb{E}_A \left(1 - \frac{p_A}{3600}\right)^k \\ &\le \left(1 - \frac{\beta\, 2^{-\dim W - r_s}}{3600}\right)^k \\ &\le \exp\left(-\frac{k\beta\, 2^{-\dim W - r_s}}{3600}\right),\end{aligned}$$

since $\operatorname{rank} A \le r_s$. Substituting $k\beta = k^{1/3}$ proves (6.3). □

6.3 The matrix-game gap

The upper bound in Lemma 6.2 tends to zero, whereas Proposition 5.3 gives a positive lower bound for the strategies extracted from a highly accepting matrix labeling.

Proposition 6.3 (A fixed gap with adjustable completeness)**.** *Fix $p_* > 0$ and choose the alphabet and shortcode data in the order of* (5.4)*. For every sufficiently large cube k, the matrix game $\mathcal{G}$ constructed from a weighted parity instance $\mathcal{E}$ with distinct positions satisfies*

$$\begin{aligned}\mathrm{OPT}(\mathcal{E}) \ge 1 - \xi \quad &\Longrightarrow \quad \mathrm{val}(\mathcal{G}) \ge 1 - k\xi - p_*/2, \\ \mathrm{OPT}(\mathcal{E}) \le 4/5 \quad &\Longrightarrow \quad \mathrm{val}(\mathcal{G}) \le 99/100.\end{aligned}$$

The lower threshold on k depends only on the fixed alphabet and shortcode data, not on $\mathcal{E}$ or ξ.

Proof. Completeness is Lemma 4.3. For soundness, fix a labeling of $\mathcal{G}$. Its restored table functions satisfy the folding and sharing identities of Section 4. Proposition 5.3 supplies a constant $\gamma > 0$ and, for every sufficiently large cube k, gives local advice strategies with agreement at least γ whenever those functions have matrix-test acceptance at least 0.99. For all sufficiently large cubes k, the upper bound in (6.3) is smaller than γ. Lemma 6.2 contradicts the existence of those strategies whenever $\mathrm{OPT}(\mathcal{E}) \le 4/5$. All thresholds are uniform over the parity instance. □

The order matters: p_* can be made arbitrarily small before the gadget parameters are fixed, and ξ can be made arbitrarily small after k is fixed. Thus the same NO gap is compatible with any prescribed small completeness error.

7 Completion of the reduction

Proposition 6.3 gives the matrix game a fixed soundness gap and an arbitrarily small completeness error. Lemma 7.1 converts it into an unweighted simple bipartite game while preserving a fixed gap. Alphabet-independent repetition, in the form of Theorem 6.1, then gives the arbitrary prescribed errors required by Theorem 1.1.

7.1 Removing weights and repeated edges

We use the following elementary construction for weighted permutation constraints. At this intermediate stage, a constraint occurrence may be a loop, and several occurrences may join the same vertices.

Lemma 7.1 (Rounding and subdivision)**.** *Let H be a finite permutation-constraint game with $h \geq 1$ positive rational occurrence weights summing to one. For every positive rational τ, put $Q = \lceil h/\tau \rceil$. There is a deterministic construction of an unweighted simple bipartite game B with $4Q$ edges and the same alphabet such that*

$$\left| \operatorname{val}(B) - \left(1 - \frac{1 - \operatorname{val}(H)}{4}\right) \right| \leq \frac{\tau}{4}. \tag{7.1}$$

If the constraints of H are translations on a finite binary vector space, so are those of B. For fixed τ, when H is listed by its vertices, occurrences, permutation tables, and binary rational weights, the construction takes polynomial time in its encoding length.

Proof. Rounding the weights. Write the occurrence weights as $w_1, \ldots, w_h$. Start with $n_e = \lfloor Q w_e \rfloor$. Increasing the $Q - \sum_e \lfloor Q w_e \rfloor$ entries with largest fractional parts by one, with ties broken in a fixed order, gives

$$n_e \geq 0, \qquad \sum_e n_e = Q, \qquad |w_e - n_e/Q| \leq 1/Q.$$

The weight differences sum to zero. Consequently, for every labeling, the change in the weight of its satisfied-occurrence set is at most

$$\frac{1}{2} \sum_{e=1}^{h} |w_e - n_e/Q| \leq \frac{h}{2Q} \leq \tau.$$

Replacing each occurrence e by n_e copies, and omitting it if $n_e = 0$, produces an unweighted multigraph H' with Q occurrences and $|\operatorname{val}(H') - \operatorname{val}(H)| \leq \tau$.

Subdivision and value. Orient each occurrence e of H' from u to v, and write its constraint as $a_v = \rho_e(a_u)$. It is replaced by the path

$$u \; -\!- p_e \; -\!- q_e \; -\!- r_e \; -\!- v,$$

using three fresh interior vertices for *every occurrence*. All original vertices and all q_e lie on the left; all p_e, r_e lie on the right. In the direction from u to v, the first three edges carry identity permutations and the last carries ρ_e. An edge expressed in left-to-right orientation carries the inverse permutation if its direction is reversed. All $4Q$ edges have equal weight.

Fix labels at the original vertices. A satisfied occurrence extends to four satisfied edges. A violated occurrence cannot extend to four, but propagating labels along the first three edges satisfies exactly three. Since the interiors are disjoint, these extensions can be chosen independently. Maximizing over the original labels gives the identity

$$\operatorname{val}(B) = 1 - \frac{1 - \operatorname{val}(H')}{4}. \tag{7.2}$$

Together with the rounding bound, this proves (7.1).

Graph form and encoding. A loop becomes a four-cycle with three distinct new vertices. Parallel occurrences have disjoint interiors. Thus each left–right vertex pair carries at most one edge, so B

is simple. An inverse binary translation is the same translation, and an identity is translation by zero; the claimed constraint form is preserved.

Finally, $Q \le h/\tau + 1$ is polynomially bounded for fixed τ. Integer division and exact comparison of rational remainders compute the n_e in polynomial time and bit complexity. Listing their Q copies and the $3Q$ new vertices is therefore polynomial as well. □

7.2 The final reduction

Proof of Theorem 1.1. Error bounds and repetition. Fix $\varepsilon, \delta \in (0, 1/2)$. Choose positive rational $\varepsilon_0 \le \varepsilon$ and $\delta_0 \le \delta$. We will take $\tau \le 1/200$ when applying the rounding and subdivision lemma. For any matrix game H satisfying the NO bound $\mathrm{val}(H) \le 99/100$, the resulting game B then satisfies

$$\mathrm{val}(B) \le 1 - \frac{1/100 - \tau}{4} \le 1 - \frac{1}{800}. \tag{7.3}$$

This gap does not depend on the alphabet. We can therefore use alphabet-independent repetition to choose its exponent before the alphabet construction fixes that alphabet. Choose an integer $t \ge 1$ such that

$$\left(1 - \frac{1}{10240000}\right)^t \le \delta_0. \tag{7.4}$$

For example, $t \ge \lceil 10240000 \log(1/\delta_0) \rceil$ suffices. By Theorem 6.1, this exponent reduces every projection game of value at most $1 - 1/800$ to value at most δ_0.

Choose positive rational parameters

$$p_* \le \frac{\varepsilon_0}{t}, \qquad \tau \le \min\left\{\frac{1}{200}, \frac{\varepsilon_0}{t}\right\}. \tag{7.5}$$

Noise and dimension parameters. Choose the rank threshold r_* from Lemma 3.1, put $\eta = 2^{-r_*}/16$, and choose the corresponding shortcode parameters α, r_s, using a positive rational lower bound for α. Then choose a sufficiently large ℓ, as specified in (5.4), and fix the corresponding alphabet data $(\mathcal{V}, K, C, \mu)$ and a splitting $\mathcal{V} = K \oplus W$, with $K = \mathbb{F}_2^\ell$.

For these rational error bounds, the choices are effective. For rational p_*, the effective assertion of Lemma 3.1 supplies all the finite alphabet data, including a computable rational noise law, by a terminating deterministic construction. The effective bounds stated after Theorem 5.1 supply the rational choice of α and the dimension thresholds, including $m_0(\eta, \ell)$ after ℓ has been fixed.

Choosing the tuple length. Next choose a cube $k \ge 8$ large enough for Proposition 6.3. Such a cube can be found by checking the estimates used in its proof. Recall that $m = 1 + 2k$ and $\beta = k^{-2/3}$; since k is a cube, β is rational. Require $m \ge m_0(\eta, \ell)$ and

$$\frac{\alpha^2}{32}\, 2^{(\ell - r_s)(m - r_s)} > (2r_s + 1)(\ell + m).$$

The latter is a sufficient rational condition for the union bound in Lemma 5.5 to be less than one. For witness transfer in Lemma 5.8, require

$$\frac{1}{4}\left[\left(1 + \frac{|\mathcal{V}| - 1}{3}\beta^2\right)^k - 1\right] < \frac{\gamma_1^2}{(1 + \alpha/4)^2},$$

where γ_1 is from (5.12). The left-hand side is the square of the total-variation bound. For the final contradiction, require

$$\left(1 - \frac{\beta\, 2^{-\dim W - r_s}}{3600}\right)^k < \gamma,$$

with γ from Proposition 5.3. All these conditions are rational and hold for sufficiently large cubes k. Enumerating cubes until they hold therefore finds a suitable k; the search cost depends only on the two prescribed errors and is incurred before the input formula is used.

Finally, with k fixed, choose a positive rational

$$\xi < \min\left\{\frac{1}{100}, \frac{\varepsilon_0}{2kt}\right\}. \tag{7.6}$$

Every choice depends only on the two prescribed errors. In particular, r_* is fixed before ℓ, the advice dimensions before k, and k before the parity error ξ.

Gap and output structure. Theorem 4.1 and Lemma 4.2 transform a given 3SAT formula into a parity instance, from which Section 4 constructs the matrix game H. Proposition 6.3 gives

$$\begin{aligned} \text{YES: } & \operatorname{val}(H) \geq 1 - d, \quad d = k\xi + p_*/2 \leq \varepsilon_0/t, \\ \text{NO: } & \operatorname{val}(H) \leq 99/100. \end{aligned}$$

Lemma 7.1 with τ gives an unweighted simple bipartite game B satisfying

$$\begin{aligned} \text{YES: } & \operatorname{val}(B) \geq 1 - (d + \tau)/4, \\ \text{NO: } & \operatorname{val}(B) \leq 1 - 1/800, \end{aligned}$$

where the NO bound is (7.3). The final output is $G = B^{\otimes t}$. Its NO value is at most $\delta_0 \leq \delta$, by (6.1) and (7.4). On a YES input, applying a labeling of B coordinatewise and using a union bound gives

$$\operatorname{val}(G) \geq 1 - \frac{t(d+\tau)}{4} \geq 1 - \varepsilon_0/2 \geq 1 - \varepsilon.$$

The output has the claimed graph and constraint form. Its two vertex sets are the t-fold products of the two sides of B. A pair of product vertices determines at most one edge in each coordinate, since B is simple, and hence at most one product edge. Independent uniform edge sampling is uniform on these product edges. The constraints of H are translations on K, by Section 4; rounding and subdivision preserve them. The product of translations with offsets $c_1, \ldots, c_t$ is translation by $(c_1, \ldots, c_t)$ on $K^t = \mathbb{F}_2^{\ell t}$. This is the alphabet $\mathbb{F}_2^s$ in the theorem, with $s = \ell t \geq 1$ fixed by the prescribed errors.

Polynomial encoding. Suppose the preprocessed parity instance has $N_{\mathcal{E}}$ positive equation occurrences. There are $N_{\mathcal{E}}^k$ equation tuples. The number of tables, noise outcomes, and perturbation vectors for each tuple is fixed after the parameter choices above, so the number h of listed matrix-test occurrences is a fixed constant times $N_{\mathcal{E}}^k$. Zero-probability outcomes are omitted, and repeated occurrences are retained separately. Each weight is a product of k parity weights, a fixed rational noise probability, and fixed uniform-sampling factors. It has polynomial binary length.

Exact table keys and folded representatives are computed by linear algebra in fixed dimensions, keeping the original variable and occurrence IDs. At most $2h$ endpoints appear. Even comparing every pair of keys to identify equal vertices takes polynomial time. The resulting distinct vertices receive consecutive numbers. Largest-remainder rounding uses $Q = \lceil h/\tau \rceil = O(h)$ copies; subdivision creates $3Q$ new vertices and $4Q$ edges. The final product has $(4Q)^t$ edges and polynomially many vertices, because t is fixed. Full permutation tables on the fixed alphabet $\mathbb{F}_2^{\ell t}$ have constant size, and endpoint indices have $O(t \log(h+1))$ bits. These observations give a deterministic polynomial-time construction and complete the proof. □

8 Approximation consequences

Theorem 1.1 turns established conditional hardness results into NP-hardness results. The underlying conditional reductions and approximation thresholds are due to the cited authors. The finite instance models below require additional arguments, given here and in Appendix A. Two further results come from companion manuscripts.

For a maximization problem, ratio r means returning value at least r times the optimum; for a minimization problem, factor c means returning value at most c times the optimum. All factors and tolerances below are fixed independently of input size. In each reduction using Theorem 1.1, they determine the Unique Games errors before its alphabet is chosen. If $\mathrm{P} \neq \mathrm{NP}$, the resulting hardness statements exclude deterministic polynomial-time approximations at the indicated strict thresholds.

8.1 Constraint satisfaction and Vertex Cover

For Max-Cut, which maximizes the number of edges crossing a bipartition, Khot, Kindler, Mossel, and O'Donnell's reduction gives NP-hardness of approximation at every fixed ratio in $(\alpha_{\mathrm{GW}}, 1)$, where the Goemans–Williamson constant is $\alpha_{\mathrm{GW}} \simeq 0.87856$ [24, Theorem 1]. This matches the semidefinite approximation threshold of Goemans and Williamson [16]. The reduction uses the Majority Is Stablest theorem of Mossel, O'Donnell, and Oleszkiewicz [32]. For Vertex Cover, which minimizes the number of vertices meeting every edge, Khot and Regev's reduction gives NP-hardness at every fixed factor in $(1, 2)$ [27].

A constraint satisfaction problem (CSP) is specified here by a finite nonempty domain D and a finite family Λ of Boolean predicates on D, of bounded arity. A weighted Λ-instance has a finite variable set and a finite collection of predicates from Λ applied to tuples of variables, with nonnegative rational weights summing to one. An assignment maps the variables to D; its value is the total satisfied weight, and OPT is the maximum of this value over all assignments. Raghavendra's theorem characterizes the optimal approximation for every fixed language of this form, up to arbitrarily small positive error, by a basic semidefinite relaxation [36, Theorem 1.1 and Corollary 1.3]. The Vertex Cover result uses its own reduction. Write SDP_R for the value of Raghavendra's relaxation, as defined in [37, Section 4.5].

The fixed-instance form of Raghavendra's theorem and the ordering theorem below use a strengthened bipartite formulation of Unique Games. In its YES case, one labeling satisfies every edge incident to at least a $1 - \zeta$ fraction of left vertices; in its NO case, every labeling satisfies at most a ν fraction of edges. These are uniform fractions of the left vertices and edges, respectively. Lemma A.1 gives this promise on an explicit nonempty simple graph whose common positive left degree can meet any fixed lower bound and be divisible by any fixed positive integer. It follows Khot and Regev's normalization and sequence construction [27, Lemmas 3.3, 3.4, and 3.6], with an additional restriction that makes the graph simple. Its error parameters and degree are fixed before the alphabet is chosen, so Theorem 1.1 supplies the required premise.

Corollary 8.1 (Raghavendra's consequence)**.** *Let J be a fixed Λ-instance with $A = \mathrm{OPT}(J) < B = \mathrm{SDP}_R(J)$. For every fixed $0 < \eta < (B - A)/2$, it is NP-hard to distinguish Λ-instances I satisfying*

$$\mathrm{OPT}(I) \geq B - \eta \qquad \textit{from those satisfying} \qquad \mathrm{OPT}(I) \leq A + \eta.$$

Proof. We first apply the fixed-instance verifier theorem, then approximate its fixed finite probability law by rational weights. Enumerating the resulting constraint occurrences gives an explicit

polynomial-size instance, and a uniform total-variation bound controls the change in every assignment's value.

The verifier gap. For the thesis's common-arity convention [37, Definitions 7.2.1–7.2.2], identify D with a fixed $[q]$, and pad every predicate with ignored trailing arguments to a common positive arity, filling the added positions with a fixed variable of J, which exists because $A < B$. This leaves each applied payoff and its dependence set unchanged, hence preserves OPT and the relaxation of [37, Section 4.5]. In that relaxation, each local distribution is over assignments to the variables on which its applied payoff actually depends. When emitting an occurrence, discard only the added trailing positions, retaining the original predicate from Λ and its weight.

Apply Raghavendra's fixed-instance theorem [37, Theorem 7.9] with tolerance $\eta/2$; see also [36, Theorem 1.1 and Lemma 2.3]. The fixed witness and tolerance determine its strengthened Unique Games errors, so the formulation above and Theorem 1.1 supply its premise, with a fixed alphabet $[M]$. Its verifier instances have optimum greater than $B-\eta/2$ in the YES case and at most $A+\eta/2$ in the NO case.

Rational approximation of the local law. We pass from the verifier's probability weights to the stated rational weights. For this fixed invocation, the construction in [37, Sections 7.3.1 and 7.5] gives a local probability law θ on a finite set S, after the alphabet $[M]$ is fixed. A record $s \in S$ specifies an applied predicate of J, its $a(s)$ ordered oracle-query strings in D^M, and the positions filled by their replies. This law may depend on J, η, M, but no varying input parameter enters it. Rational probability vectors are dense in this finite simplex, so fix $\widehat{\theta} \in \mathbb{Q}^S_{\geq 0}$ with

$$\sum_{s\in S} \widehat{\theta}(s) = 1, \qquad \delta := \frac{1}{2}\sum_{s\in S} |\widehat{\theta}(s) - \theta(s)| < \frac{\eta}{2}.$$

This rational table is fixed finite data of the reduction; its description length may depend on J, η, M.

The finite occurrence list. Let L_Φ and V_Φ be the left and right sides of the nonempty simple game supplied above, and let $h \geq 1$ be its common left degree. The variable list is $V_\Phi \times D^M$, as in Section 7.5 of the cited thesis. Its outer verifier chooses a uniform left vertex u and, independently for each oracle query, chooses a uniform neighbor of u and relabels the query string by that edge's permutation. For each $s \in S$, $u \in L_\Phi$, and ordered neighbor tuple in $N(u)^{a(s)}$, the output contains the Λ-predicate prescribed by s on the resulting query variables. All positions of any syntactic variable omitted from the test are filled with the first listed output variable; the applied predicate is independent of such variables, so this leaves its value unchanged. The occurrence has weight

$$\frac{\widehat{\theta}(s)}{|L_\Phi| h^{a(s)}}.$$

These weights are nonnegative rationals summing to one. Keeping equal occurrences separate gives $|L_\Phi| \sum_{s\in S} h^{a(s)}$ occurrences and $|V_\Phi||D|^M$ variables. The local set and every $a(s)$ are fixed, while the graph has polynomial size; hence this is a polynomial occurrence list. The explicit neighbor lists and edge permutations let us write it in polynomial time. Each weight has polynomial binary length: it can be represented with a fixed denominator multiplied by $|L_\Phi| h^{a(s)}$.

Preservation of the gap. For an assignment σ, let $g_\sigma(s) \in [0,1]$ be its conditional acceptance probability over these outer choices. The original and rationalized instances use the same variable set, and

$$|\mathrm{val}_{\widehat{\theta}}(\sigma) - \mathrm{val}_\theta(\sigma)| = \left|\sum_{s\in S} (\widehat{\theta}(s) - \theta(s)) g_\sigma(s)\right| \leq \delta < \frac{\eta}{2}.$$

Taking maxima gives the same bound for their optima. The imported gap therefore yields optimum at least $B-\eta$ in the YES case and at most $A+\eta$ in the NO case, as required. □

8.2 Ordering constraints

An ordering CSP uses a global order of its input variables; the size of this variable set is part of the input. Fix an integer $k \geq 2$ and a nonempty proper subset $\Pi \subset S_k$, where S_k is the set of permutations of $\{1, \ldots, k\}$. In the finite probability-weighted model of Guruswami, Håstad, Manokaran, Raghavendra, and Charikar [17, Definition 8.2], we use an explicitly listed finite variable set V and a finite list of constraint occurrences. Their nonnegative rational weights are encoded by binary numerators and denominators and sum to one. Every occurrence is an ordered k-tuple $(v_1, \ldots, v_k)$ of distinct variables, satisfied by a linear order $\prec$ when

$$v_{\pi(1)} \prec \cdots \prec v_{\pi(k)} \qquad \text{for some } \pi \in \Pi.$$

Write $\mathrm{OPT}_\Pi(I)$ for the maximum satisfied weight. A uniform random order satisfies expected weight $\rho_\Pi = |\Pi|/k!$. Since $\mathrm{OPT}_\Pi(I) \leq 1$, it is a ρ_Π-approximation. For fixed k, conditional expectation derandomizes this guarantee in polynomial time with exact rational arithmetic; the complete calculation and bit-length bound appear at the end of Section A.5.

Corollary 8.2 (Ordering-CSP consequence)**.** *Fix $k \geq 2$ and $\varnothing \neq \Pi \subsetneq S_k$, and put $\rho_\Pi = |\Pi|/k!$. For every fixed $0 < \varepsilon < (1 - \rho_\Pi)/2$, it is NP-hard to distinguish finite weighted Π-ordering-CSP instances I satisfying*

$$\mathrm{OPT}_\Pi(I) \geq 1 - \varepsilon \qquad \textit{from those satisfying} \qquad \mathrm{OPT}_\Pi(I) \leq \rho_\Pi + \varepsilon.$$

Consequently, approximation at every fixed ratio $r \in (\rho_\Pi, 1]$ is NP-hard.

Remark. Together with the random-ordering guarantee, this gives the optimal threshold ρ_Π.

Proof. Proposition A.2 constructs the stated finite instance from a simple Unique Games instance with the promise of Lemma A.1. It uses the local gap and analytic estimates of Guruswami et al. [17] and fixes the required game errors and degree conditions before the alphabet is chosen. Lemma A.1 supplies that game from Theorem 1.1; composing the reductions proves the displayed gap for the original predicate Π.

For a fixed $r \in (\rho_\Pi, 1]$, choose the positive error still smaller so that $\varepsilon < (r - \rho_\Pi)/(1 + r)$. Then $r(1 - \varepsilon) > \rho_\Pi + \varepsilon$, so an r-approximation would distinguish the two cases. □

For Maximum Acyclic Subgraph in the finite weighted directed-graph model, take $k = 2$ and $\Pi = \{12\}$ in one-line notation. A constraint (u, v) asks that the edge point forward; forward edges form an acyclic subgraph, and every acyclic subgraph has a topological order. Its threshold is therefore $1/2$. For Betweenness, the constraint (a, b, c) asks that b lie between a and c, so $\Pi = \{123, 321\} \subset S_3$ and the threshold is $2/3! = 1/3$ [17, Theorem 1.1 and Section 1.2]. These finite weighted consequences have near-perfect completeness with fixed positive error.

8.3 Cut, deletion, and clustering consequences

For the cut problems of Chawla, Krauthgamer, Kumar, Rabani, and Sivakumar [9, Section 1], let $G = (V, E)$ be a finite undirected graph with positive rational edge costs c_e, and let $\mathcal{D} = \{\{s_i, t_i\} : 1 \leq i \leq k\}$ contain $k \geq 1$ distinct demand pairs with $s_i \neq t_i$. For $M \subseteq E$, write $c(M) = \sum_{e \in M} c_e$, and let $q_{\mathcal{D}}(M)$ count the demand pairs disconnected in $G - M$. Weighted Multicut minimizes $c(M)$ subject to $q_{\mathcal{D}}(M) = k$. Their nonuniform Sparsest Cut search problem minimizes $c(M)/q_{\mathcal{D}}(M)$ over cutsets M with $q_{\mathcal{D}}(M) > 0$. Only the selected pairs carry unit demand; they need not be all vertex pairs.

For weighted Min-2CNF$^{\equiv}$ Deletion, the input consists of clauses $(\ell \equiv \ell')$ with positive rational weights, where ℓ, ℓ' are Boolean literals, and an assignment pays the total weight of false clauses. The allowed clauses therefore express equality or disequality of variables.

In the arbitrary weighted-graph form of Correlation Clustering, each present edge has a $+$ or $-$ label and a nonnegative rational weight. A partition into any number of clusters pays for $+$ edges whose endpoints lie in different clusters and $-$ edges whose endpoints lie in the same cluster. Missing edges carry no penalty. This is the minimum-disagreements objective of [10, Section 2.1].

Corollary 8.3 (Cut, deletion, and clustering hardness)**.** *For every fixed real constant $C > 1$, it is NP-hard to approximate weighted* Multicut*, nonuniform* Sparsest Cut*, weighted* Min-2CNF$^{\equiv}$ Deletion*, and* Correlation Clustering *on arbitrary weighted graphs within factor C. For* Sparsest Cut*, this is hardness of the search task under polynomial-time Cook reductions: the algorithm must output a cutset, not only an estimate of the optimum. Consequently, unless* P = NP*, none of these problems admits a deterministic polynomial-time approximation with any fixed constant factor.*

Proof. Corollary 1.3 of [9] gives the first three conclusions under its ordinary edge-value Unique Games hypothesis. For a requested factor C, use its hardness statement at any fixed larger factor L: a C-approximation would also be an L-approximation. To supply that hypothesis, orient the bipartite constraints of Theorem 1.1 left-to-right, renumber its fixed alphabet as $[d]$ with $d = 2^s \geq 2$, and give its edges equal weights summing to one. Padding the smaller vertex side with isolated questions if needed preserves both value and polynomial size. The search-version qualification for Sparsest Cut is in Section 1.3 of the cited source.

For Correlation Clustering, the hard Multicut instances in [9, Section 2.1] have distinct antipodal demand pairs inside d-cubes, while the edges of the Multicut graph are cube edges or join different cubes. Here $d \geq 2$, unchanged by Lemma 1.5 of that source, so the demand pairs are nonedges. The transformation of Demaine, Emanuel, Fiat, and Immorlica [10, Theorem 4.7 and Corollary 4.8] therefore creates no opposite-signed parallel edges. It preserves the optimum and converts every clustering into a multicut of no greater cost, so the same fixed-factor hardness transfers. □

The separate manuscript [33, Theorem 1.1] proves NP-hardness at every fixed factor $C > 1$ for the bipartition objective with nonnegative rational capacities and exactly unit demand between every pair of distinct vertices. This uniform-demand theorem does not depend on Theorem 1.1 and is not supplied by the nonuniform reduction above.

8.4 Kernel clustering

A separate companion application concerns the following maximization problem. For *identity-target kernel clustering*, fix an integer $k \geq 3$. An input is an explicitly represented rational symmetric matrix $A = (a_{pq})_{p,q=1}^{N} \succeq 0$ with $A\mathbf{1} = 0$, where $\mathbf{1}$ is the all-ones vector. An assignment $\sigma : \{1, \ldots, N\} \to \{1, \ldots, k\}$ has value

$$\mathrm{val}_A(\sigma) = \sum_{p,q=1}^{N} a_{pq} \mathbf{1}_{\{\sigma(p)=\sigma(q)\}}, \qquad \mathrm{OPT}_k(A) = \max_{\sigma} \mathrm{val}_A(\sigma).$$

The sum includes ordered pairs and the diagonal; empty clusters are allowed, and there is no normalization by cluster size. A loss factor $\alpha \geq 1$ means returning value at least $\mathrm{OPT}_k(A)/\alpha$. The

Gaussian propeller companion uses Theorem 1.1 to prove NP-hardness of achieving every fixed loss factor

$$1 \le \alpha < \alpha_k, \qquad \alpha_k = \frac{8\pi}{9}\left(1 - \frac{1}{k}\right),$$

for each fixed $k \ge 3$ [35, Theorem 6.1]. This is strict hardness below the propeller rounding factor α_k.

A Finite formulations of the consequences

Some of the reductions in Section 8 use a bipartite Unique Games instance in which a nearly satisfying labeling satisfies all constraints incident to most left vertices. Lemma A.1 obtains a simple graph with this property and with a prescribed lower bound on its common left degree. Proposition A.2 then gives the finite rational formulation of the ordering consequence. Its local construction follows Guruswami, Håstad, Manokaran, Raghavendra, and Charikar [17]; the composition below converts it to constraints on distinct variables.

A.1 Strong satisfaction in simple bipartite games

Orient the constraints of a bipartite game from its left side to its right side. A left vertex is *strongly satisfied* by a labeling when every edge incident to it is satisfied.

Lemma A.1 (Strong satisfaction in simple games)**.** *Fix $0 < \zeta, \nu < 1$ and integers $d, H \ge 1$. There exist an integer $h \ge H$ with $d \mid h$ and errors $\zeta_0, \gamma_0 \in (0, 1/2)$, depending only on ζ, ν, d, H, with the following property. Applying Theorem 1.1 with errors ζ_0, γ_0 supplies an alphabet $[R]$, $R \ge 1$, for which there is a deterministic polynomial-time reduction from 3SAT to nonempty explicit unweighted simple bipartite Unique Games instances $\Phi_\varphi = (A, B, E)$ with the following properties. Every left vertex has degree h. If φ is satisfiable, one labeling strongly satisfies at least a $1 - \zeta$ fraction of A. If φ is unsatisfiable, then $\mathrm{val}(\Phi_\varphi) < \nu$.*

Proof. We adapt the normalization and sequence construction of Khot and Regev [27, Lemmas 3.3, 3.4, and 3.6]. We first normalize and round the edge weights, then form left vertices from sequences of edge occurrences. Extra copies of right vertices allow us to retain sequences with distinct neighbors, making the final graph simple.

Parameter choices. Choose a multiple h of d with

$$h \ge \max\{H, 2\}, \qquad \frac{1}{h} < \frac{\nu}{4}.$$

For $(T)_h = T(T-1)\cdots(T-h+1)$, choose a fixed integer $T \ge h$ such that

$$p_{\mathrm{graph}} := \frac{(T)_h}{T^h} > \frac{3}{4}.$$

Such a choice exists because $1 - (T)_h/T^h \le \binom{h}{2}/T$. Choose positive rationals β, γ, u and a positive integer L_2 so that

$$\beta < \frac{\nu}{4}, \qquad h^2\gamma < \frac{\nu}{4}, \qquad (h+1)u < \frac{\zeta}{2}, \qquad \frac{1}{L_2} < \min\left\{\frac{\beta\gamma}{2}, \frac{\zeta}{2h}\right\}.$$

Finally choose positive rational $\gamma_0, \zeta_0 < 1/2$ with

$$2\gamma_0 < \frac{\beta\gamma}{2}, \qquad \sqrt{2\zeta_0} < u.$$

All these choices precede the invocation of Theorem 1.1 with completeness error ζ_0 and soundness error γ_0.

Normalizing the left weights. Write the output of Theorem 1.1 as $\Phi_0 = (X_0, Y, E_0)$, orienting each permutation left-to-right and renumbering its alphabet as $[R]$. Put $M_0 = |E_0| > 0$, give each edge weight $1/M_0$, and let w_x be the total incident weight at x. Simplicity gives one permutation at each pair of positive weight. Create

$$n_x = \lfloor 2|X_0|w_x \rfloor$$

copies of x, each with local edge weights w_{xy}/w_x. Only vertices with $w_x > 0$ have copies. If X_1 is the copy set, then

$$|X_0| \le |X_1| = \sum_x n_x \le 2|X_0|,$$

and the incident weight at each copy is one.

Here and below a local satisfied weight uses the labels at that left vertex and at the shared right vertices. Given any labeling of the copy game, keep the right labels and give all copies of each x a label that maximizes their common local objective. The corresponding labeling L of Φ_0 satisfies

$$\sum_{x' \in X_1} w_{\mathrm{sat}}(x') \le \sum_{x : n_x > 0} \frac{n_x}{w_x} w_{\mathrm{sat},L}(x) \le 2|X_0|\, w_{\mathrm{sat},L}(\Phi_0).$$

Consequently the copy game's normalized value is at most $2\gamma_0$ on a NO input. On a YES input, lift a labeling of Φ_0 with unsatisfied weight at most ζ_0. The same calculation for unsatisfied weight gives mean local loss at most $2\zeta_0 < u^2$. Markov's inequality leaves more than a $1 - u$ fraction of X_1 with local satisfied weight at least $1 - u$.

Rounding to edge occurrences. Put $\alpha = L_2|Y|$. At each $x \in X_1$, write $\widetilde{w}_{xy}$ for its normalized local weights. Choose one neighbor y_0 of positive local weight, round $\alpha\widetilde{w}_{xy}$ down for every $y \ne y_0$, and assign the remaining occurrences to y_0. All occurrences at a pair retain that pair's permutation. This gives an unweighted multigraph Φ_{occ} of left degree α. For any subset of the neighbors, the difference between its old weight and its new occurrence fraction has magnitude at most $|Y|/\alpha = 1/L_2$. This follows by summing the rounding losses over the subset if it omits y_0, and over its complement otherwise. This applies to the satisfied subset for every labeling. Hence

$$\mathrm{val}(\Phi_{\mathrm{occ}}) \le 2\gamma_0 + \frac{1}{L_2} < \beta\gamma \qquad \text{on a NO input.} \tag{A.1}$$

On a YES input, the labeling above has satisfied occurrence fraction at least $1 - a$, where $a = u + 1/L_2$, at more than a $1 - u$ fraction of X_1.

Copies of right vertices. Index the α occurrences at x by j, with right endpoint y_j and permutation π_j. Replace each $y \in Y$ by (y, t), $t \in [T]$, and replace occurrence j by all T occurrences from x to (y_j, t) with permutation π_j. Call the resulting left-αT-regular multigraph Φ_{tag}. For any labeling of Φ_{tag}, keep its left labels and, independently for each original y, give y the label of a uniformly chosen copy (y, t). The expected edge value in Φ_{occ}, counting its occurrences, equals the given edge value in Φ_{tag}. Thus

$$\mathrm{val}(\Phi_{\mathrm{tag}}) \le \mathrm{val}(\Phi_{\mathrm{occ}}) < \beta\gamma \qquad \text{on a NO input.} \tag{A.2}$$

The tag-independent lift of the YES labeling preserves every local satisfied fraction.

Sequence construction and soundness. Consider the full sequence graph whose left vertices are

$$(x, (j_1, t_1), \ldots, (j_h, t_h)), \qquad j_i \in [\alpha], \quad t_i \in [T],$$

with position i joined to (y_{j_i}, t_i) by π_{j_i}. This is the sequence construction of Khot and Regev; its left vertex identifier includes the entire ordered occurrence sequence. The left label may depend on that entire sequence. Each x contributes $(\alpha T)^h$ full sequence vertices, all of degree h. Thus the uniform edge value is the average satisfied fraction over a uniform x, an ordered h-tuple of independent uniform occurrences at x, and a uniform position in that tuple.

Fix arbitrary right labels in Φ_{tag}. Each occurrence at x determines the unique left label that satisfies it. Let $p_x(\ell)$ be the fraction determining label ℓ. The $p_x(\ell)$ sum to one, and choosing the best left label at each x gives

$$\mathbb{E}_x \max_{\ell} p_x(\ell) \le \mathrm{val}(\Phi_{\mathrm{tag}}) < \beta\gamma.$$

Fewer than a β fraction of x therefore have $\max_\ell p_x(\ell) \ge \gamma$. At every other x, the probability that two labels determined by h independent occurrences agree is at most

$$\binom{h}{2} \sum_{\ell} p_x(\ell)^2 \le \binom{h}{2} \max_{\ell} p_x(\ell) < h^2\gamma.$$

If no two determined labels agree, any label of the sequence vertex satisfies at most one of its h edges. On the remaining sequences its satisfied fraction is at most one. Averaging first over sequences and then over x, and using $1/h + h^2\gamma < 1$, bounds the full sequence graph's value by

$$s := \beta + \frac{1}{h} + (1-\beta)h^2\gamma < \frac{3\nu}{4}. \tag{A.3}$$

The estimate holds for every choice of the right labels and every sequence-dependent choice of left labels.

Restriction to distinct neighbors. Retain only sequence vertices for which $t_1, \dots, t_h$ are pairwise distinct. For each x and each base sequence $(j_1, \dots, j_h)$, exactly $(T)_h$ tag sequences remain. Each x thus contributes $\alpha^h(T)_h$ retained vertices, a fraction p_{graph} of its full set. The h right neighbors of each retained vertex are distinct because their tags are distinct. Different occurrence sequences remain different left vertices, so this retained graph is simple and has common left degree h.

Give each sequence vertex based at x the old label of x, and keep the tag-independent right labels. For this lifted YES labeling, satisfaction at position i depends only on j_i. Conditioning the tags to be distinct leaves the base occurrence sequence independent and uniform. If the satisfied fraction at x in Φ_{occ} is f_x, exactly an f_x^h fraction of its retained sequence vertices are strongly satisfied. Since every x contributes equally, the strongly satisfied fraction is at least

$$(1-u)(1-a)^h \ge 1 - u - ha = 1 - (h+1)u - \frac{h}{L_2} > 1 - \zeta.$$

Here $0 < a < 1$ follows from the parameter choices.

For NO, extend any labeling of the retained graph to the full graph by giving discarded left vertices arbitrary labels and keeping all right labels. All left degrees are h, so its full edge value is at least p_{graph} times its retained edge value. By (A.3), the latter is at most

$$\frac{s}{p_{\mathrm{graph}}} < \frac{3\nu/4}{3/4} = \nu.$$

This extension permits arbitrary dependence of the retained left labels on their tags.

Polynomial encoding. All the operations are finite and deterministic. If N bounds the explicit size of Φ_0, then $|X_1| = O(N)$ and $\alpha = L_2|Y| = O(N)$. The retained graph has $|X_1|\alpha^h(T)_h = O(N^{h+1})$ left vertices and h edges per left vertex. Its identifiers and the rational arithmetic used for rounding have polynomial bit length, and its permutation tables use the fixed alphabet inherited from Φ_0. This proves the claimed polynomial bound and completes the reduction. □

A.2 Ordering predicates and the finite transfer

The ordering theorem of Guruswami et al. uses a multiscale local instance and a low-influence rounding estimate. Their general outer composition is described as analogous to the two-query construction in their Section 7. The following proposition gives the composition for the rational occurrence-list model of Section 8. It separates the constants chosen for the ordering gap from the subsequently chosen Unique Games alphabet.

Proposition A.2 (Finite ordering transfer)**.** *Fix $k \geq 2$ and $\emptyset \neq \Pi \subsetneq S_k$, put $\rho_\Pi = |\Pi|/k!$, and fix $0 < \delta < (1 - \rho_\Pi)/2$. There are integers $q, H \geq 1$ and positive rationals $\zeta, \nu < 1$, depending only on k, Π, δ, with the following property for every fixed integer $R \geq 1$.*

Let $\Phi = (A, B, E)$ be a nonempty explicit unweighted simple bipartite Unique Games instance over $[R]$, oriented from A to B, whose common left degree h satisfies $h \geq H$ and $q \mid h$. There is a deterministic transformation, polynomial in the explicit size of Φ for fixed R, producing an explicit list of Π-ordering constraints on distinct variables within each constraint, with nonnegative rational weights, each encoded by a binary numerator and a positive binary denominator, summing to one. If some labeling strongly satisfies at least $(1 - \zeta)|A|$ left vertices, then $\mathrm{OPT}_\Pi(I) \geq 1 - \delta$. If $\mathrm{val}(\Phi) \leq \nu$, then $\mathrm{OPT}_\Pi(I) \leq \rho_\Pi + \delta$. The constants q, H, ζ, ν are independent of both R and h.

The proof combines a finite local ordering instance with the input game. Lemma A.3 supplies the local gap and rational distributions. To bound the payoff of a global order, we divide each local order into q consecutive blocks and average block membership over neighbors. Lemma A.5 bounds the payoff of these block ranks when the resulting functions have small influences; Lemma A.4 bounds the probability of a shared block, where coarsening can change the induced order. Large influences yield labels for the input game. The composition in Section A.5 finishes by conditioning on distinct neighbors and refining the completeness rank map to a strict order.

For $P = \mathbf{1}_\Pi$, choose $\pi_0 \in \Pi$ and set

$$P_0(\sigma) = P(\pi_0 \circ \sigma), \qquad \sigma \in S_k.$$

Then $P_0(\mathrm{id}) = 1$, and its average is ρ_Π. For a vector of integer ranks $r = (r_1, \ldots, r_k)$, let $\mathrm{Ref}(r)$ be the permutations compatible with their weak order: they list positions in increasing rank and may order positions of equal rank arbitrarily. Define

$$\overline{P}_0(r) = \frac{1}{|\mathrm{Ref}(r)|} \sum_{\sigma \in \mathrm{Ref}(r)} P_0(\sigma). \tag{A.4}$$

This is the uniform tie extension used in [17, Section 8.2]. It lies in $[0, 1]$ and agrees with P_0 when the ranks are distinct.

The normalization preserves the predicate after one permutation of tuple positions. Given a raw tuple $w = (w_1, \ldots, w_k)$, define the emitted tuple v by

$$v_{\pi_0(j)} = w_j \qquad (j \in [k]). \tag{A.5}$$

If a strict order induces σ on w, it induces $\pi_0 \circ \sigma$ on v. More generally, relabeling positions gives the bijection

$$\mathrm{Ref}\big(f(v_1), \ldots, f(v_k)\big) = \{\pi_0 \circ \sigma : \sigma \in \mathrm{Ref}(f(w_1), \ldots, f(w_k))\}$$

for every rank map f. Uniform averages over these sets therefore give equal P- and P_0-payoffs, including ties. The permutation of positions in (A.5) preserves distinctness and weights.

A.3 Local ordering data

For a finite local instance $J = (U, \mathcal{C}, \omega)$, write $t_c = (u_{c,1}, \dots, u_{c,k})$ for the tuple of occurrence $c \in \mathcal{C}$ and ω_c for its probability weight. Its q-ordering value is

$$\mathrm{opt}_q(J) = \max_{g:U\to[q]} \sum_{c\in\mathcal{C}} \omega_c \overline{P}_0(g(u_{c,1}), \dots, g(u_{c,k})).$$

Lemma A.3 (Rational local ordering data)**.** *Fix $k \geq 2$, $\varnothing \neq \Pi \subsetneq S_k$, and $\pi_0 \in \Pi$ as in Section A.2; use $\rho_\Pi = |\Pi|/k!$ and the tie extension* (A.4)*. For every integer $q \geq 1$ and rational $\eta > 0$, there are a finite local instance $J = (U, \mathcal{C}, \omega)$, with $|U| = m \geq 2$, and probability laws μ_c on $[m]^k$ with the following properties. Every tuple c has k distinct variables; the weights ω_c and all atoms of μ_c are rational. Every one-position marginal of μ_c is uniform on $[m]$, and there is a fixed $0 < \alpha \leq 1/2$ such that $\mu_c(x) > \alpha$ for every c, x. Moreover*

$$\mathrm{opt}_q(J) \leq \rho_\Pi + \eta, \qquad v_{\mathrm{loc}} := \sum_c \omega_c \mathbb{E}_{X\sim\mu_c} \overline{P}_0(X) \geq 1 - \eta. \tag{A.6}$$

The local laws have common feasible Gram data for the relaxation in [17, Section 8.3]: there are a unit vector I and vectors $b_{u,j}$, $u \in U$, $j \in [m]$, with

$$\sum_j b_{u,j} = I, \qquad \langle b_{u,i}, b_{u,j}\rangle = 0 \quad (i \neq j), \qquad \|b_{u,j}\|^2 = \frac{1}{m},$$

and, for every two positions r, s of c,

$$\langle b_{u_{c,r},i}, b_{u_{c,s},j}\rangle = \Pr_{X\sim\mu_c}[X_r = i, X_s = j]. \tag{A.7}$$

All these data are fixed by k, Π, q, η.

Proof. The explicit increasing-tuple construction in [17, Theorem 11.1, Definition 11.2, and the proof of Theorem 2.5], applied to P_0, gives a finite instance J^* with $\mathrm{opt}_q(J^*) \leq \rho_\Pi + \eta$ and a linear order satisfying every constraint. Its distribution uses only uniform choices from finite sets, so its weights are rational. Its tuple positions lie in successive disjoint intervals and are therefore distinct.

The finite-copy smoothing from the proof of [17, Lemma 8.9], specialized to this complete order, supplies uniform marginals while preserving the gap. Take L disjoint copies of J^*, each of weight $1/L$, and place their complete orders consecutively in an order $\sigma : U \to [m]$, where $m = L|U^*|$. The resulting instance J still has $\mathrm{opt}_q(J) \leq \rho_\Pi + \eta$, since the value of any q-assignment is the average of its values on the copies. Choose L and a positive rational $s_0 < 1$ so that $1/L + s_0 < \eta$. Define a random assignment $X : U \to [m]$ as follows: with probability $1 - s_0$, choose a uniform cyclic shift of the ranks of σ; with probability s_0, assign the variables independently and uniformly from $[m]$.

Each variable is uniform in either part of the mixture. A cyclic cut can split at most one of the L consecutive copies, so every shift has payoff at least $1 - 1/L$. The mixture's payoff is therefore at least $(1 - s_0)(1 - 1/L) > 1 - \eta$. Let μ_c be its restriction to the variables of c. If $n_c(x)$ is the number of the m cyclic shifts with restriction x, then

$$\mu_c(x) = (1 - s_0)\frac{n_c(x)}{m} + \frac{s_0}{m^k}.$$

These atoms are rational and exceed $\alpha := s_0/(2m^k)$. The denominator m^k is valid because the variables in c are distinct.

Finally regard the finite mixture as a probability space and take $I = 1$ and $b_{u,j} = \mathbf{1}_{\{X(u)=j\}}$ in its L^2 space. Uniform marginals, disjoint events for different labels of one variable, and restrictions to each c give all the displayed Gram identities. This constructs the required feasible data for this particular local instance. □

A.4 Low-influence estimates

For scalar functions on the uniform product $[m]^R$, use

$$\mathrm{Inf}_\ell(F) = \mathbb{E}[\mathrm{Var}_{z_\ell}(F \mid z_{-\ell})].$$

The noise operator T_θ, $0 \le \theta \le 1$, independently keeps each coordinate with probability θ and otherwise resamples it uniformly, then averages the function. In an orthonormal product expansion these are

$$T_\theta F = \sum_\sigma \theta^{|\sigma|} \widehat{F}(\sigma)\chi_\sigma, \qquad \mathrm{Inf}_\ell(F) = \sum_{\sigma:\sigma_\ell \neq 0} \widehat{F}(\sigma)^2. \tag{A.8}$$

In particular, further smoothing decreases every influence.

Lemma A.4 (Uniform collision estimate)**.** *For every $0 < \kappa < 1/2$, there is $\mu_0(\kappa) > 0$. For each $m \ge 2$, there is a nonnegative function $r_c^{m,\kappa}(t) \to 0$ as $t \downarrow 0$ with the following property, uniformly over all $R \ge 1$. Let $F, G : [m]^R \to [0,1]$ have the same mean $0 < \mu \le \mu_0(\kappa)$ and satisfy*

$$\mathrm{Inf}_\ell(T_{1-\kappa}F) \le t, \qquad \mathrm{Inf}_\ell(T_{1-\kappa}G) \le t \qquad (\ell \in [R]).$$

For every coupling (Z, Z') whose two marginals are uniform on $[m]^R$,

$$\mathbb{E}\big[(T_{1-2\kappa}F)(Z)(T_{1-2\kappa}G)(Z')\big] \le \mu^{1+\kappa/2} + r_c^{m,\kappa}(t). \tag{A.9}$$

The function $r_c^{m,\kappa}$ is independent of R, μ, F, G and the coupling.

Proof. This is the dimension-uniform form of [17, Lemmas 3.4 and 3.5] needed here. For $H = T_{1-\kappa}F$, the Fourier formula gives

$$\|T_{1-2\kappa}F\|_2^2 \le \mathrm{Stab}_{1-\kappa}(H),$$

where $\mathrm{Stab}_\theta(H) = \sum_\sigma \theta^{|\sigma|}\widehat{H}(\sigma)^2$. Indeed, $(1-\kappa)^3 - (1-2\kappa)^2 = \kappa(1-\kappa-\kappa^2) > 0$. Each full coordinate influence bound for H implies the corresponding low-degree influence bound in Theorem 4.4 of Mossel, O'Donnell, and Oleszkiewicz [32]. With product dimension R, one-coordinate atom bound $1/m$, and correlation $1-\kappa$, that theorem gives, for all sufficiently small t,

$$\mathrm{Stab}_{1-\kappa}(H) \le \Gamma_{1-\kappa}(\mu) + C\frac{\log(2m)}{\kappa}\frac{\log\log(1/t)}{\log(1/t)}.$$

The cutoff and this error are independent of R and μ. Here $\Gamma_\theta(\mu)$ is the Gaussian halfspace noise stability of a set of Gaussian measure μ.

For fixed κ, Theorem B.5 of the same source gives $\Gamma_{1-\kappa}(\mu) = \mu^{2/(2-\kappa)+o(1)}$ as $\mu \downarrow 0$. Since

$$\frac{2}{2-\kappa} - \left(1 + \frac{\kappa}{2}\right) = \frac{\kappa^2}{2(2-\kappa)} > 0,$$

there is a cutoff $\mu_0(\kappa) > 0$, independent of m, R, below which $\Gamma_{1-\kappa}(\mu) \le \mu^{1+\kappa/2}$. This proves the corresponding squared L^2 bound for both F and G, with an error $r_c^{m,\kappa}(t) \to 0$ independent of the stated variables. Choose a cutoff $t_0 < e^{-e}$ within the range of the displayed bound, use its nonnegative error for $0 < t < t_0$, and set $r_c^{m,\kappa}(t) = 1$ for $t \ge t_0$. The latter range follows from the product being at most one. Cauchy–Schwarz under any coupling with the specified marginals proves (A.9). □

Write $\mathcal{S}_q = \{p \in [0,1]^q : \sum_j p_j = 1\}$ and define the multilinear extension of $\overline{P}_0$ on $[q]^k$ by

$$P_0^{\mathrm{ml}}(p_1, \dots, p_k) = \sum_{r \in [q]^k} \overline{P}_0(r) \prod_{i=1}^{k} p_i^{r_i}, \qquad p_i \in \mathcal{S}_q. \tag{A.10}$$

Lemma A.5 (A simplex-valued local estimate)**.** *Let J, μ_c, α be the data of Lemma A.3, and fix $0 < \lambda < 1$. For an integer $R \geq 1$, choose c with probability ω_c. Conditional on this one choice of c, choose the R coordinate tuples independently from μ_c, producing strings $Z_1, \dots, Z_k \in [m]^R$. Then obtain $\widetilde{Z}_i$ by independently resampling every position and coordinate uniformly with probability λ. For a single function $F = (F^1, \dots, F^q) : [m]^R \to \mathcal{S}_q$, used for every c and every position, suppose that $0 < t < 1/(kq)$ and*

$$\mathrm{Inf}_\ell(T_{1-\lambda} F^j) \leq t \qquad (\ell \in [R],\ j \in [q]).$$

Then

$$\mathbb{E}_{c,Z,\widetilde{Z}} P_0^{\mathrm{ml}}\big(F(\widetilde{Z}_1), \dots, F(\widetilde{Z}_k)\big) \leq \mathrm{opt}_q(J) + r_s(kqt), \tag{A.11}$$

where $r_s(u) \to 0$ as $u \downarrow 0$, independently of R. The function r_s depends only on the fixed local data, k, q, λ.

Proof. We prove the simplex-valued estimate of [17, Lemma 10.5] by comparing the smoothed functions with a common Gaussian ensemble, then rounding that ensemble to an assignment of J. The comparison requires the scalar influence and tail bounds verified below.

Smoothing and matched ensembles. Put $H = T_{1-\lambda} F$. Independence of the noise between positions and multilinearity turn the left side of (A.11) into

$$\mathbb{E}_c \mathbb{E}_{Z \sim \mu_c^{\otimes R}} P_0^{\mathrm{ml}}\big(H(Z_1), \dots, H(Z_k)\big). \tag{A.12}$$

For each coordinate, the integral ensemble consists of the indicators $\mathbf{1}_{\{X_i = j\}}$ under μ_c; its atoms have probability greater than α. The common Gram data (A.7) supply one Gaussian ensemble for all variables of J with the same first and second moments on every constraint, as in [17, Lemma 10.4]. Explicitly, for a standard Gaussian vector g in the span of the Gram vectors, the variables

$$G_{u,j} = \langle I, b_{u,j} \rangle + \langle b_{u,j} - \langle I, b_{u,j} \rangle I, g \rangle$$

have those moments. Take independent copies of this ensemble in the R coordinates. Each H^j is evaluated on these ensembles through its multilinear product expansion. Equivalently, an orthonormal basis on $[m]$ can be expressed as linear combinations of the symbol indicators, and the same polynomial is used for the integral and Gaussian ensembles.

Influence and tail bounds. For a fixed c, let $\mathcal{T}_\theta^c$ retain each whole coordinate tuple with probability θ and otherwise resample it from μ_c. Uniform one-position marginals give, componentwise,

$$\mathcal{T}_{1-\lambda}^c [F^j(Z_i)] = (T_{1-\lambda} F^j)(Z_i) = H^j(Z_i).$$

Thus these are the smoothed components required by the invariance theorem on the product of the tuple ensembles. The same marginals give the influence identity below; the superscript indicates the product measure used to compute influence:

$$\mathrm{Inf}_\ell^{\mu_c^{\otimes R}}(H^j(Z_i)) = \mathrm{Inf}_\ell^{[m]^R}(H^j). \tag{A.13}$$

Indeed, conditioning on every coordinate tuple except the ℓth leaves the ℓth symbol at position i uniform, and the other symbols at that position have the uniform product law. For a vector, influence is the sum of its component influences [21, Definition 2.6]. Thus the vector $(H^j(Z_i))_{i \in [k], j \in [q]}$ has each coordinate influence at most kqt; each component has variance at most one.

For the remaining tail hypothesis, let $(F^j(Z_i))_S$, $S \subseteq [R]$, be the Hoeffding components on the tuple product $\mu_c^{\otimes R}$, and let the superscript $> d$ retain the terms with $|S| > d$. The tuple-noise identity above multiplies component S by $(1-\lambda)^{|S|}$. Since the unsmoothed F^j takes values in $[0,1]$, for every real $d > 0$,

$$\begin{aligned}\operatorname{Var}\big((H^j(Z_i))^{>d}\big) &= \sum_{|S|>d} (1-\lambda)^{2|S|} \big\|(F^j(Z_i))_S\big\|_2^2 \\ &\le (1-\lambda)^{2d} \operatorname{Var}(F^j(Z_i)) \le (1-\lambda)^{2d}.\end{aligned}$$

All norms and variances here use $\mu_c^{\otimes R}$. This supplies the tail bound in [21, Theorem 3.6]. These are sufficient hypotheses for the vector-valued invariance estimate stated in [17, Theorem 10.1].

Comparing the payoffs. Let $\mathbf{p} : \mathbb{R}^q \to \mathcal{S}_q$ be Euclidean projection. It is nonexpansive. The function

$$\Psi(x_1, \dots, x_k) = P_0^{\mathrm{ml}}(\mathbf{p}(x_1), \dots, \mathbf{p}(x_k))$$

is Lipschitz with constant at most $\sqrt{kq}$: on the simplex, telescoping the positions bounds the payoff difference by $\sum_i \|x_i - y_i\|_1$, and projection and Cauchy–Schwarz give the claim. For this function, the cited invariance estimate applies separately for each c, with vector dimension kq and influence threshold kqt, and gives error at most

$$r_s(kqt), \qquad r_s(u) = C_{kq}\sqrt{kq}\, u^{\lambda/(18\log(1/\alpha))}.$$

In particular the error is independent of R.

Rounding to an assignment. On the integral ensemble, $H(Z_i) \in \mathcal{S}_q$, so Ψ equals the payoff in (A.12). On the common Gaussian ensemble, project $H(G_u)$ to $\mathcal{S}_q$ for every $u \in U$, and independently draw one q-label for each u with those probabilities. This is one random assignment of J. The variables in every local tuple are distinct, so its conditional expected payoff at that tuple is the multilinear payoff Ψ. Every realized assignment has value at most $\mathrm{opt}_q(J)$. Averaging the invariance comparisons over c proves (A.11). □

A.5 Composition and finite encoding

Proof of Proposition A.2. The independent-neighbor experiment gives the completeness and soundness estimates. Conditioning on distinct neighbors then produces valid ordering constraints, with the change in payoff controlled by the probability of a repeated neighbor. We derive these estimates for every R before selecting their numerical parameters independently of R.

The local experiment. Fix for now $0 < \kappa < 1/2$, q with $1/q \le \mu_0(\kappa)$, and local data from Lemma A.3. Use the local experiment of Lemma A.5 with resampling probability $\lambda = 2\kappa$. Also fix auxiliary constants $0 < t < 1/(kq)$ and $\chi > 0$.

For an input graph Φ, let $\pi_{a\to b}$ denote the permutation on edge ab. Its action on strings is fixed by

$$(\pi_{a\to b} z)_{\pi_{a\to b}(\ell)} = z_\ell \qquad (\ell \in [R]).$$

The output variable set will be $\mathcal{V} = B \times [m]^R$. Consider the following independent neighbor experiment: choose a uniformly from A, choose $b_1, \dots, b_k$ independently and uniformly from $N(a)$, and independently perform the local experiment, obtaining $\widetilde{Z}_1, \dots, \widetilde{Z}_k$. Its raw output positions are

$$w_i = (b_i, \pi_{a\to b_i} \widetilde{Z}_i) \in \mathcal{V}. \tag{A.14}$$

For any rank map $f : \mathcal{V} \to \mathbb{Z}$, define

$$V(f) = \mathbb{E}\overline{P}_0(f(w_1), \dots, f(w_k)), \qquad V_a(f) = \mathbb{E}[\overline{P}_0(f(w_1), \dots, f(w_k)) \mid a].$$

The bounded payoff (A.4) is used even if the experiment repeats an output ID.

Completeness. Suppose a labeling L strongly satisfies at least $1-\zeta$ of the left vertices. Define the coarse rank map $f(b,z) = z_{L(b)}$. At every strongly satisfied a and every neighbor b,

$$f(b, \pi_{a\to b} z) = z_{\pi_{a\to b}^{-1}(L(b))} = z_{L(a)}.$$

Thus all k positions read the same label coordinate, for every choice of neighbors. With probability $(1-2\kappa)^k$ none of these k symbols is resampled; conditional on that event their tuple has law μ_c. Nonnegativity of the payoff gives

$$V(f) \geq (1-\zeta)(1-2\kappa)^k v_{\mathrm{loc}}. \tag{A.15}$$

Soundness for small influences. Fix an injective order $O : \mathcal{V} \to [|\mathcal{V}|]$. At a fixed a, its local set $N(a) \times [m]^R$ has $M = hm^R$ distinct IDs. Sort this set by O and divide the sorted list into q consecutive blocks $C_{a,1}, \ldots, C_{a,q}$ of size M/q. These sizes are integers because $q \mid h$. Each block is the intersection of the local set with an interval of global ranks. Define

$$F_a^j(z) = \frac{1}{h} \sum_{b\in N(a)} \mathbf{1}_{\{(b,\pi_{a\to b} z)\in C_{a,j}\}}, \qquad F_a = (F_a^1, \ldots, F_a^q). \tag{A.16}$$

For every z, $F_a(z) \in \mathcal{S}_q$. Each edge permutation is a bijection on strings, so

$$\mathbb{E}_z F_a^j(z) = \frac{|C_{a,j}|}{hm^R} = \frac{1}{q}. \tag{A.17}$$

For any interval I of integer ranks put

$$F_a^I(z) = \frac{1}{h} \sum_{b\in N(a)} \mathbf{1}_{\{O(b,\pi_{a\to b} z)\in I\}}.$$

Call *a pseudorandom at threshold* t if $\mathrm{Inf}_\ell(T_{1-\kappa} F_a^I) \leq t$ for every I and every $\ell \in [R]$. At such an a, this bound holds for each block function F_a^j.

Let $j_a(w)$ be the block index of a local ID w. Conditional on the local query strings, independence of the neighbor choices gives

$$\mathbb{E}_{b_1,\ldots,b_k} \overline{P}_0\big(j_a(w_1), \ldots, j_a(w_k)\big) = P_0^{\mathrm{ml}}\big(F_a(\widetilde{Z}_1), \ldots, F_a(\widetilde{Z}_k)\big). \tag{A.18}$$

The identity includes outcomes with equal neighbors. Further smoothing decreases the block influences, so $\mathrm{Inf}_\ell(T_{1-2\kappa} F_a^j) \leq t$. Lemma A.5, applied to the single function F_a, bounds the expected coarse payoff in (A.18) by $\mathrm{opt}_q(J) + r_s(kqt)$.

When the k coarse ranks are distinct, their induced permutation is the one induced by the fine ranks $O(w_i)$. Otherwise the two payoffs differ by at most one. This also covers a repeated output ID, which necessarily produces a coarse tie. For two positions $i \neq i'$, independence of their noise and neighbor choices gives their coarse tie probability as

$$\sum_{j=1}^{q} \mathbb{E}_c \mathbb{E}_{Z\sim\mu_c^{\otimes R}} \big[(T_{1-2\kappa} F_a^j)(Z_i)(T_{1-2\kappa} F_a^j)(Z_{i'})\big].$$

Conditional on c, both strings have uniform product marginals. Lemma A.4 permits their correlation and, with the mean $1/q$ from (A.17), bounds the last display by $q^{-\kappa/2} + q r_c^{m,\kappa}(t)$. A

union bound over pairs, combined with the coarse-payoff bound from Lemma A.5, gives, at every pseudorandom a,

$$V_a(O) \le S_{\mathrm{loc}} := \mathrm{opt}_q(J) + \binom{k}{2}\left(q^{-\kappa/2} + q r_c^{m,\kappa}(t)\right) + r_s(kqt). \tag{A.19}$$

Decoding large influences. An excess over this bound yields a labeling of Φ. For each $b \in B$, restrict O to $\{b\} \times [m]^R$ and compress its ranks to obtain a strict order o_b of $[m]^R$. Let

$$\mathcal{L}_b = \{\ell \in [R] : \mathrm{Inf}_\ell(T_{1-\kappa}\mathbf{1}_{\{o_b(z)\in I\}}) > t/2 \text{ for some rank interval } I\}.$$

For every R, the few-influential-coordinates lemma [17, Definition 4.2 and Lemma 4.3] gives

$$|\mathcal{L}_b| \le \frac{400}{\kappa(t/2)^3} = \frac{3200}{\kappa t^3}. \tag{A.20}$$

If a is not pseudorandom, choose one witnessing interval I_a and coordinate ℓ_a with $\mathrm{Inf}_{\ell_a}(T_{1-\kappa}F_a^{I_a}) > t$. Noise commutes with coordinate permutations and averaging, and influence is a convex squared norm. Hence

$$\mathrm{Inf}_{\ell_a}(T_{1-\kappa}F_a^{I_a}) \le \frac{1}{h}\sum_{b\in N(a)} \mathrm{Inf}_{\pi_{a\to b}(\ell_a)}(T_{1-\kappa}\mathbf{1}_{\{O(b,z)\in I_a\}}). \tag{A.21}$$

Each term is at most one. An average greater than t forces more than a $t/2$ fraction of the terms to exceed $t/2$. The restriction of a global-rank interval to one fiber $\{b\} \times [m]^R$ is an interval of its relative order, so these neighbors satisfy $\pi_{a\to b}(\ell_a) \in \mathcal{L}_b$.

Give each such a label ℓ_a, and label every other left vertex arbitrarily. Independently at each right vertex with $\mathcal{L}_b \neq \emptyset$, choose one uniform label from $\mathcal{L}_b$; use any label when the set is empty. By (A.20), the expected satisfied fraction among the edges at each nonpseudorandom a is greater than

$$\frac{t}{2}\frac{\kappa t^3}{3200} = \frac{\kappa t^4}{6400}.$$

One random right labeling is used for all left vertices. Linearity of expectation and the common left degree identify the resulting average with the ordinary uniform edge fraction.

If $V(O) > S_{\mathrm{loc}} + \chi$, more than a χ fraction of left vertices are nonpseudorandom: if that fraction is g, the bound $0 \le V_a(O) \le 1$ and (A.19) give $V(O) \le (1-g)S_{\mathrm{loc}} + g \le S_{\mathrm{loc}} + g$. The preceding decoder would then give a labeling of edge value greater than $\chi\kappa t^4/6400$. Consequently

$$\mathrm{val}(\Phi) \le \nu < \frac{\chi\kappa t^4}{6400} \quad\Longrightarrow\quad V(O) \le S_{\mathrm{loc}} + \chi \quad \text{for every injective } O. \tag{A.22}$$

Constraints on distinct variables. Let D be the event that $b_1, \dots, b_k$ are distinct. For $h \ge k$,

$$p_D := \Pr(D) = \frac{(h)_k}{h^k}, \qquad \delta_h := 1 - p_D \le \frac{\binom{k}{2}}{h}. \tag{A.23}$$

The probability is the same at every a, and D is independent of the local experiment. Conditional on D, a remains uniform and its ordered distinct neighbor tuple is uniform among its $(h)_k$ possibilities. All output IDs are then distinct because their first coordinates b_i differ. Let I_0 denote this finite weighted distribution of strict P_0 constraints, and write $\mathrm{OPT}_{P_0}(I_0)$ for its maximum strict payoff.

Let $V_D(f)$ denote the expected tie-extended payoff after this conditioning, and use $V_{\neg D}(f)$ for conditioning on its complement. For any rank map f, coarse or injective,

$$V(f) = p_D V_D(f) + (1-p_D)V_{\neg D}(f), \qquad |V_D(f) - V(f)| \le \delta_h, \tag{A.24}$$

because all payoffs lie in $[0,1]$. For the coarse map in (A.15), this bounds the change in payoff under conditioning. Within each finite fiber of that map, choose one uniform random permutation of the distinct IDs, independently between fibers, and concatenate the fibers in rank order. On every conditioned tuple, restriction of these permutations to its IDs is uniform within each tied rank class. Thus the expected strict P_0-payoff of this one random global order equals $V_D(f)$. Some global order attains at least that value. For soundness, an injective O has distinct ranks on every conditioned tuple, so its tie-extended payoff there is its strict P_0-payoff. The completeness bound (A.15) and soundness bound (A.22) each change by at most δ_h under (A.24), giving

$$\begin{aligned} \text{YES}: &\quad \mathrm{OPT}_{P_0}(I_0) \ge (1-\zeta)(1-2\kappa)^k v_{\mathrm{loc}} - \delta_h, \\ \text{NO}: &\quad \mathrm{OPT}_{P_0}(I_0) \le S_{\mathrm{loc}} + \chi + \delta_h \qquad \text{if } \nu < \chi\kappa t^4/6400. \end{aligned} \tag{A.25}$$

These bounds are for the conditioned P_0 tuples. Applying (A.5) makes every occurrence use the original predicate Π with exactly the same values. Call this emitted instance I.

Parameter choices. For the prescribed δ, choose a positive rational η with $10\eta < \delta$, and a rational $0 < \kappa < 1/2$ with $2\kappa k < \eta$. Put $K = \binom{k}{2}$. Choose an integer q so large that

$$\frac{1}{q} \le \mu_0(\kappa), \qquad Kq^{-\kappa/2} < \eta.$$

Choose the data J, m, α from Lemma A.3 at error η. The two analytic error functions now have fixed parameters and are uniform in R. Choose a positive rational $t < 1/(kq)$ so small that

$$Kqr_c^{m,\kappa}(t) < \eta, \qquad r_s(kqt) < \eta.$$

Set $\chi = \eta$. Choose positive rationals $\zeta, \nu < 1$ with

$$\zeta < \eta, \qquad \nu < \frac{\eta\kappa t^4}{6400},$$

and choose an integer $H \ge \max\{k, q\}$ with $K/H < \eta$. These choices depend only on k, Π, δ. In particular, they precede both the choice of h in Lemma A.1 and the alphabet supplied by Theorem 1.1.

For every $h \ge H$, $\delta_h < \eta$. The local gap (A.6), Bernoulli's inequality, and (A.25) give

$$\text{YES}: \quad \mathrm{OPT}_{\Pi}(I) \ge 1 - \zeta - 2\kappa k - \eta - \delta_h > 1 - 4\eta > 1 - \delta,$$

whereas (A.19) gives

$$\text{NO}: \quad \mathrm{OPT}_{\Pi}(I) \le S_{\mathrm{loc}} + \chi + \delta_h < \rho_{\Pi} + 6\eta < \rho_{\Pi} + \delta.$$

This proves the numerical claims simultaneously for every R.

Finite encoding. Once R is fixed, the choice of c, the product $\mu_c^{\otimes R}$, and the independent rational 2κ-noise form one fixed finite rational law θ_R on local outcomes ξ. The output enumerates these local outcomes; the Gaussian ensemble serves only in the analytic comparison. Renumber B by $[|B|]$ and explicitly list $\mathcal{V} = B \times [m]^R$. For each $a \in A$, each ordered distinct neighbor tuple, and each $\xi \in \operatorname{supp} \theta_R$, list the tuple in (A.14), permuted by (A.5), with weight

$$\frac{\theta_R(\xi)}{|A|(h)_k}. \tag{A.26}$$

The weights sum to one and are nonnegative rationals. Every listed tuple has distinct IDs; the same tuple may occur in several entries. There are $|B|m^R$ variables and at most $|A|(h)_k|\operatorname{supp}\theta_R|$

entries. For fixed k, m, R, θ_R, these counts are polynomial in the explicit graph size, even when h varies. A fixed common denominator for θ_R shows that each weight has $O(\log |A| + k \log h)$ bits plus a fixed constant. The IDs have $O(\log |B|)$ bits plus a fixed constant. Exact enumeration of these entries is therefore deterministic and polynomial for fixed R, as asserted. □

The same occurrence-list encoding also justifies the deterministic random-order guarantee in Section 8. Build an order by exposing one next variable at a time. If exactly s variables of a constraint have been exposed, its conditional success probability is the number of permutations in Π whose first s positions are the roles of those variables in the exposed order, divided by $(k - s)!$. It can be computed by inspecting the fixed set S_k. For each choice of the next variable, sum these probabilities against the listed rational weights and choose a next variable whose conditional expectation is at least the average over the choices. This preserves the initial expectation $|\Pi|/k!$. All comparisons are exact and polynomial in the input bit length: the product of the weight denominators has bit length at most their total encoded length, and the factorial denominators have constant size for fixed k. The variable list is explicit, so the number of prefix steps and candidate choices is polynomial as well.

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